Do Patient Metadata Still Help Strong Image Models? Representation-Dependent Attenuation of Auxiliary Information in Skin Lesion Classification

Authors to be confirmed
Affiliation to be confirmed

Abstract—Fusing patient metadata with lesion images is widely reported to improve skin cancer classification. We ask whether this benefit is a property of the data or of a weak image model. We define the added value of metadata as the out-of-fold reduction in cross-entropy when metadata are added to an image score, and relate its population counterpart, $I(Y; A \mid S)$, exactly to information the image cannot carry plus a representation-dependent residual. In a pre-registered analysis on HAM10000, added value fell from 18.6 to 1.7×10^{-3} nats from colour histograms to ConvNeXt-T. In two pre-registered replication waves covering 6 further datasets, this attenuation held in all 6 after Holm correction and survived a strictly cross-fitted estimator, three fusion estimators, complete cases and every missing-data encoding, whereas attenuation between ResNet-18 and ConvNeXt-T depended on the analysis. A pre-registered intervention locates the mechanism for age: removing the linear trace of age from the image representation raised age's added value beyond performance-matched random and sham erasure on a held-out confirmatory dataset (Holm $p = 0.002$) and three replication datasets, and was specific to the erased variable; it was uninformative on the three smallest datasets. The remaining benefit was concentrated: the quarter of lesions with the largest expected gain carried at least 70% of it, whereas a pre-registered nearest-neighbour graph of the representation added no predictive information. The non-redundant share of the metadata's information ranged from 0.01 to 0.50 and depended on the label definition. We also document three evaluation pitfalls: stacking with shared folds leaks test labels, missing-value encoding can halve or triple added value, and circular-shift saliency tests are invalid for centred lesions.

Index Terms—skin lesion classification, patient metadata, multimodal fusion, conditional mutual information, representation learning, pre-registration, cross-fitting, concept erasure

I. Introduction

Clinicians do not diagnose skin lesions from images alone: age, anatomical site, and lesion history all shape the assessment. Following this intuition, many automated systems fuse patient metadata with image features. A systematic review of such systems found that every included study reported a gain from patient data, while noting that it remains unclear how each variable contributes [1]. On the smartphone dataset PAD-UFES-20, adding clinical data improved balanced accuracy by about 7% across several networks [2].

A reported gain, however, conflates two things: information in the metadata that the image genuinely lacks, and information the image contains but a particular model failed to extract. A patient's age, for example, is partly visible in the skin around a lesion. If an image model is weak, metadata can compensate for what it misses; if it is strong, the same metadata may be redundant. Whether metadata "help" may therefore depend on the image model they are added to, a dependence that single-backbone studies cannot reveal.

This paper makes five contributions:

1) An interpretation of added value. Using the chain rule of mutual information, we write the Bayes added value of metadata A to a representation $S = g(X)$ of an image X as $I(Y; A \mid S) = I(Y; A \mid X) + R(S)$, where $I(Y; A \mid X)$ is the information unavailable from the image and the residual $R(S)$ vanishes as $S \to X$ (Proposition 1). The decomposition is elementary; its value is that it says what a measured gain from metadata does and does not show. It does not imply that added value falls as representations improve; it does predict that removing from S what it shares with A restores A's value, which we test by intervention.
2) A measurement protocol. We estimate added value out of fold along a ladder of frozen image representations, with grouped cross-validation, inner-fold tuning, strict cross-fitting of the two stages, and grouped bootstrap inference, and we audit metadata for label leakage, including label-informative missingness.
3) Pre-registered evidence on seven datasets. A pre-registered analysis on HAM10000 [3] and two pre-registered replication waves: ISIC 2020 [4], ISIC 2019 excluding HAM10000 [5] and PAD-UFES-20 [6]; then HIBA [7], MSK 2020 [8] and MILK10k [9], whose paired dermoscopic and clinical images of the same lesions allow a test of imaging modality. Two pre-registered mechanism tests failed, and we report why.

4) Interventions on the representation. Fine-tuning, training-set size, backbone size within a family, a 304M-parameter vision transformer, and pre-registered linear concept erasure with matched random and sham controls separate what attenuates metadata value (a better image model for the task; linearly decodable image features that encode the variable) from what does not (more parameters on frozen features).
5) Where metadata help, and evaluation pitfalls. Benefit concentrates in lesions the image model finds uncertain and can be ranked without labels, whereas a representation graph adds nothing beyond the fitted models (pre-registered). Stacking with shared folds leaks test labels, missing-value encoding, image- vs. patient-weighting and the fusion estimator can change conclusions, and the circular-shift null used for saliency–mask overlap is invalid when lesions are centred.

II. Related Work

Metadata fusion in dermatology. Concatenating one-hot metadata with image features is the dominant approach, and all eleven studies in a systematic review reported a gain [1]. The closest prior evidence is a benchmark of nine backbones on an ISIC subset [10]: averaged over its backbones, metadata raised AUROC by 0.023 for models trained from scratch but by only 0.001 for ImageNet-pretrained, fine-tuned models (our calculation from its Table 5; single runs, no intervals). The authors did not frame this as attenuation, and to our knowledge no study has measured added value along a controlled ladder of image models with grouped inference and pre-specified tests. A reader study on MILK10k found dermoscopy and clinical close-ups diagnostically complementary [11]; whether the modality changes what metadata add to an automated model has not been tested.

Incremental value. In clinical prediction, the added value of a new variable is commonly tested with nested models; comparing nested-model AUCs with DeLong's test is invalid [12]. In logistic models, omitting a predictor that is independent of a covariate still attenuates that covariate's coefficient (non-collapsibility) [13]. This mechanism makes directly acting metadata gain value as the image model improves (Section III).

Per-sample utility and concept erasure. The expected per-sample reduction in log-loss from observing a feature equals its pointwise conditional mutual information, and dynamic feature selection trains networks to predict it [14], [15]; we use the same quantity to ask where metadata help, not to select features. Linear concept erasure removes all linearly decodable information about a concept from a representation in closed form [16]; we use it as an intervention on the image representation. Amnesic probing removes a property from a representation and compares the change in the model's own behaviour with removing the same number of random directions [17]; we instead measure the change in the added value of a separately observed copy of the property, the metadata, against random subspaces matched for performance cost, sham erasers and an information dose. Removal guided by a concept probe can also remove task-relevant features when the concept is correlated with the label [18], a failure we observe on our two most age-dependent datasets.

Leakage and shortcuts. Splitting medical images without grouping by subject inflates accuracy substantially [19]. Dermatology datasets contain several images per lesion and per patient, and acquisition source can predict diagnosis.

Spatial nulls. Comparisons between spatial maps must preserve spatial autocorrelation in the null distribution; naive voxel permutation is anti-conservative [20], [21]. We carry this principle over to the evaluation of saliency maps against anatomy.

III. Theory

Let X be an image, A metadata, $Y \in \{0, 1\}$ the label, and $S = g(X)$ a representation. With log-loss $\ell(y, p) = -\log p(y)$, the Bayes predictors satisfy $\mathbb{E}\,\ell(Y, P(Y \mid S)) = H(Y \mid S)$, so the expected reduction in log-loss from adding A is

$$\Delta(S) = H(Y \mid S) - H(Y \mid S, A) = I(Y; A \mid S). \tag{1}$$

Proposition 1. For any measurable g and $S = g(X)$,

$$I(Y; A \mid S) = I(Y; A \mid X) + \underbrace{I(Y; X \mid S) - I(Y; X \mid S, A)}_{R(S)}. \tag{2}$$

If g is injective with a measurable inverse, $R(S) = 0$. If moreover $Y \perp\!\!\!\perp A \mid X$, then $\Delta(S) = 0$.

Proof. Expand $I(Y; A, X \mid S)$ with the chain rule in two orders [22]: $I(Y; A \mid S) + I(Y; X \mid S, A) = I(Y; X \mid S) + I(Y; A \mid X, S)$. Because S is a function of X, conditioning on (X, S) equals conditioning on X, so $I(Y; A \mid X, S) = I(Y; A \mid X)$. Rearranging gives (2). If g is injective, X is a function of S, so both terms of $R(S)$ vanish. The final claim follows because $I(Y; A \mid X) = 0$ when $Y \perp\!\!\!\perp A \mid X$. □

Interpretation. $I(Y; A \mid X)$ is the part of the metadata's information that no image model can recover. $R(S)$ measures how the representation's losses interact with A. $R(S) > 0$ when A substitutes for image information that S discarded, and $R(S) < 0$ indicates synergy, the information-theoretic face of non-collapsibility [13]. The identity does not order representations: for nested $S_1 = h(S_2)$, $\Delta(S_1) - \Delta(S_2) = R(S_1) - R(S_2)$ can take either sign (if $X = A = Y$ is one fair bit, refining S from a constant to X removes one bit of added value; if $Y = X \oplus A$ for independent fair bits, it creates one), and the frozen representations we compare are not nested. What it does predict is the effect of an intervention:

removing from S the information it shares with A should raise Δ when that shared information is what made A redundant (Section V-H). Injective Borel maps between standard Borel spaces have measurable inverses, which covers images and feature vectors. Proposition 1 was checked numerically to machine precision on 200 random discrete distributions and coarsenings (code release).

Estimation. We estimate Δ for each rung k of a capacity ladder by a two-stage out-of-fold procedure. First, an image score $s_k(x)$ is obtained from frozen features $f_k(x)$ by L_2-regularised logistic regression, with the penalty chosen by grouped inner cross-validation within each training fold. Second, logistic models on $[s_k]$ and $[s_k, a]$ are fitted and evaluated out of fold:

$$\hat{\Delta}_k = \frac{1}{n}\sum_{i=1}^{n}\Big[\ell\big(y_i, \hat{p}_0^{(-i)}\big) - \ell\big(y_i, \hat{p}_1^{(-i)}\big)\Big], \tag{3}$$

where $(-i)$ denotes a model fitted without the fold containing i. In the registered analyses the second stage was trained on the out-of-fold scores of the other folds, which came from first-stage models that had seen the test fold; Section IV describes a strictly cross-fitted version, used for every analysis added after the audit. For fixed fitted models q_0, q_1, the expected held-out reduction equals $I(Y; A \mid s_k) + \mathrm{KL}(p_{Y|s} \,\|\, q_0) - \mathrm{KL}(p_{Y|s,A} \,\|\, q_1)$, so $\hat{\Delta}_k$ estimates conditional mutual information only for well-specified second stages; its target conditions on the one-dimensional score s_k, which agrees with $I(Y; A \mid S_k)$ only if s_k is sufficient for Y given S_k with and without A; and it can be negative in finite samples. $\hat{\Delta}_k$ averages over images, so lesions or patients with many images weigh more. Uncertainty is quantified by a bootstrap over grouping units (lesions or patients) with the fitted models held fixed, which omits training variability [23], and differences between rungs by a paired grouped bootstrap of per-image loss differences.

Where metadata help. With $u(s) = I(Y; A \mid S = s)$, $\Delta(S) = \mathbb{E}\, u(S)$ and $u(s) \le H(Y \mid S = s)$: metadata can help only where the image leaves the label uncertain. For a calibrated fusion model, the expected per-image benefit given (s, a) is $K = \mathrm{KL}(q_1 \| q_0)$, which needs no label [15]. Writing $\Delta(S) = H(Y \mid S)\,\eta(S)$ with $\eta \in [0, 1]$ splits attenuation into the image resolving more cases (H falls) and metadata resolving a smaller share of what remains (η falls).

IV. Experimental Setup

Datasets (Table I). HAM10000 (dermoscopy; 10,015 images of 7,470 lesions) [3]; ISIC 2020 (dermoscopy; 33,126 images of 2,056 patients) [4]; ISIC 2019 with all HAM10000 lesions removed [5]; PAD-UFES-20 (smartphone images; 2,298 images of 1,373 patients) [6]; the dermoscopic images of HIBA [7]; MSK 2020, consecutive biopsies from one centre [8]; and MILK10k, in which every lesion has one dermoscopic and one clinical close-up image [9]. The two MILK10k image sets are analysed separately but are not independent. Among datasets that carry ISIC identifiers, no image appears in two datasets; PAD-UFES-20 has no such identifiers.

Leakage audit. Fields that encode how the diagnosis was established were excluded: HAM10000's dx_type (alone, AUROC 0.76 for melanoma) and PAD-UFES-20's biopsed (every cancer is biopsied). On HAM10000 the image source alone predicts melanoma (AUROC 0.73), and frozen deep features identify the source almost perfectly (AUROC up to 0.995), so we report source-adjusted analyses. A separate audit of missingness found that it, too, can carry the label. In PAD-UFES-20, all 804 images lacking gender, skin type, and history fields are benign and unbiopsied, so missingness is a proxy for the excluded biopsed field, and "unknown" growth and change histories are 88% malignant. Smaller associations exist elsewhere (e.g. ISIC 2020: no melanoma among the 65 images of unknown sex). As a further control, we refit the second stage after permuting the metadata between grouping units (50 permutations). Because the pre-registered encoding keeps missing-value indicators, we add a complete-case sensitivity analysis (Section V-F).

Ladder. HSV colour histograms (48-d); ImageNet-pretrained ResNet-18 and ResNet-50 [24], ConvNeXt-T [25], and ViT-B/16 [26]; frozen, pooled features from 224×224 centre crops.

Protocol. 5-fold stratified grouped cross-validation (lesion or patient) with three fold assignments; inner grouped 3-fold CV for the penalty; 2,000 grouped bootstrap resamples. The HAM10000 analysis and both replication waves were pre-registered, with hypotheses, the Holm correction [27] and its family (six tests in wave 1, eight in wave 2) fixed before the respective data were modelled. Wave 2 restricted HIBA to dermoscopic images in a dated addendum written after reading its metadata and before any modelling. Additions, all exploratory and labelled as such: the complete-case analysis (prompted by the missingness audit), the retained fraction ρ, the source-adjusted and paired HAM10000 analyses, the interpretability analyses, the representation-strength manipulations, and the recoverability, permutation, and cross-dataset analyses; the last group was described in a dated registry before implementation. Two implementation errors found during the work (an image-name mismatch that dropped ISIC 2019 images, and a missing-value check that missed NaN) were fixed before any affected result was used. A later audit found that the second stage had been trained on scores from first-stage models that saw the outer test fold. The strictly cross-fitted estimator trains both second-stage models on inner grouped out-of-fold scores computed within each training fold, scores the test fold with the first stage fitted on the whole training fold, and imputes metadata with training-fold statistics. Called unchanged, the original code reproduced all 120 stored results; the corrected estimator changed them by at most 0.42 of a bootstrap CI half-width (HAM10000, ConvNeXt-T: 1.7

TABLE I

Datasets. Positives: melanoma for HAM10000 and ISIC; malignant otherwise. Groups define the cross-validation and bootstrap units. *Images without a lesion identifier form their own group. †No lesion identifier is available. The two MILK10k rows show the same lesions.

Dataset	Modality	Images	Groups	Pos.	Metadata
HAM10000	dermoscopy	10,015	7,470 lesions	1,113	age, sex, site
ISIC 2020	dermoscopy	33,126	2,056 patients	584	age, sex, site
ISIC 2019 (non-HAM)	dermoscopy	15,316	6,461 lesions*	3,409	age, sex, site
PAD-UFES-20	smartphone	2,298	1,373 patients	1,089	demogr. + symptoms
HIBA	dermoscopy	1,224	563 patients	533	age, sex, site, skin type
MSK 2020	mixed	1,174	1,174 images†	148	age, sex, site
MILK10k (derm.)	dermoscopy	5,117	5,117 lesions	3,634	age, sex, site
MILK10k (clin.)	clin. close-up	5,117	5,117 lesions	3,634	age, sex, site

TABLE II

HAM10000, melanoma vs. rest, lesion-grouped 5-fold CV. AUROC and $\hat{\Delta}$ are means over three fold assignments; the CI is the 95% lesion bootstrap for assignment 0. Recoverability: out-of-fold R^2 for age and mean AUROC for the five commonest sites and for image source, predicted from the frozen features.

Representation	Image AUROC	$\hat{\Delta}$ (10^{-3} nats)	95% CI (10^{-3})	Recoverability age	site	source
HSV histogram	0.765	18.6	[12.2, 24.3]	0.06	0.65	0.921
ResNet-18	0.853	3.7	[−0.1, 8.1]	0.28	0.73	0.979
ResNet-50	0.850	3.0	[−0.5, 6.9]	0.32	0.73	0.989
ConvNeXt-T	0.889	1.7	[−2.7, 5.6]	0.39	0.79	0.995
ViT-B/16	0.887	2.2	[−1.1, 6.3]	0.37	0.79	0.995

TABLE III

HAM10000 robustness (fold assignment 0). ΔAUROC and ΔBrier (10^{-3}) from adding metadata, with 95% lesion-bootstrap CIs. Last two columns: out-of-fold log-loss of score + metadata (two-stage, as in Table II) vs. all frozen features + metadata in one model; similar values mean the one-dimensional score loses little.

Representation	ΔAUROC [95% CI]	ΔBrier [95% CI]	Log-loss with metadata two-stage	joint
HSV histogram	+0.043 [+0.030, +0.056]	4.84 [3.02, 6.70]	0.288	0.285
ResNet-18	+0.007 [+0.001, +0.013]	1.29 [0.23, 2.42]	0.253	0.250
ResNet-50	+0.005 [−0.001, +0.011]	1.16 [0.14, 2.23]	0.255	0.255
ConvNeXt-T	+0.003 [−0.002, +0.007]	0.88 [−0.14, 1.93]	0.231	0.228
ViT-B/16	+0.004 [−0.001, +0.008]	0.79 [−0.25, 1.86]	0.230	0.228

TABLE IV

HAM10000: paired differences in added value between representations, $\hat{\Delta}_a - \hat{\Delta}_b$ (10^{-3} nats, 95% paired lesion-bootstrap CI), fold assignment 0. Exploratory: not corrected for multiplicity.

Comparison	Unadjusted	Source-adjusted
HSV histogram − ResNet-18	14.3 [10.7, 18.0]	10.6 [7.9, 13.7]
HSV histogram − ConvNeXt-T	16.7 [12.6, 20.6]	13.2 [9.6, 16.7]
ResNet-18 − ConvNeXt-T	2.4 [0.7, 4.0]	2.6 [0.9, 4.0]
ResNet-18 − ViT-B/16	1.4 [−0.1, 2.9]	1.5 [−0.1, 3.0]
ResNet-50 − ConvNeXt-T	1.6 [−0.1, 3.2]	1.8 [0.2, 3.4]

to 2.4×10^{-3} nats). Registered results are reported as computed; every analysis added after the audit (E40–E55) uses the corrected estimator and was described in a dated registry before implementation. The erasure experiments (E50–E55) were registered with a confirmatory family of four tests (Holm) on three datasets never used in an earlier erasure experiment (ISIC 2020, HIBA, MSK 2020); the other datasets serve as replications.

V. Results

A. Pre-registered analysis on HAM10000

Fig. 1 and Table II show that metadata reduce out-of-fold log-loss by 18.6×10^{-3} nats for colour histograms, with a CI excluding zero in all three fold assignments. For the deep representations the reduction is only 3.7×10^{-3} to 1.7×10^{-3} nats, and the log-loss CIs include zero. Table III expresses the same effect in familiar units: the AUROC gain falls from 0.043 [0.030, 0.056] for colour histograms to 0.007 [0.001, 0.013] for ResNet-18, which remains detectable, and to 0.003 [−0.002, 0.007] for ConvNeXt-T, which does not. Fitting all frozen features and metadata in one model gives nearly the same log-loss as the two-stage design (Table III), so compressing the image to a one-dimensional score does not hide an interaction.

Recoverability of age from the image rises along the ladder (R^2 from 0.06 to 0.39), consistent with Proposition 1 for metadata whose information the image carries. A per-variable analysis did not sharpen this: age, sex, and site retained 14%, 22%, and 18% of their colour-histogram added value at the strongest rungs, and even sex proved recoverable from dermoscopy (AUROC 0.77). Because all three variables became recoverable together, HAM10000 alone cannot isolate the mechanism, which motivates the symptom analysis on PAD-UFES-20 below. Three of four pre-registered predictions held; the predicted non-zero floor was not observed.

B. Shortcut control and paired tests

With image source included in both models, added value still falls (from 15.3×10^{-3} to 2.1×10^{-3} nats). Within single sources, melanoma AUROC rises with capacity, so the deep representations do read lesions and do not merely detect the source. The paired attenuation from ResNet-18 to ConvNeXt-T is 2.4×10^{-3} nats (95% CI $[0.7, 4.0] \times 10^{-3}$). Across all 8 image sets, the real metadata outperformed every one of 50 group-level permutations at ConvNeXt-T on complete cases (empirical $p = 0.02$, the smallest attainable), so the residual added value is not an artefact of fitting extra parameters.

C. Pre-registered replication

Table V and Fig. 1 report both waves. After Holm correction within each registered family, H1 (colour histogram vs. ConvNeXt-T) held in 3 of 3 datasets in wave 1 and 4 of 4 image sets in wave 2, that is, in all 6 independent datasets (MILK10k counted once, and as holding only if both of its image sets held). H2 (ResNet-18 vs. ConvNeXt-T) held in 1 of 3 and 3 of 4 (MSK 2020: $p_{\text{Holm}} = 0.086$), i.e. in 3 of 6 independent datasets. With the strictly cross-fitted estimator, H1 held in 6 and H2 in 4 of 6: ISIC 2020's H2 moved from $p_{\text{Holm}} = 0.092$ to 0.028 on a difference of 0.36×10^{-3} nats. Across fold assignments, patient weighting, alternative label definitions, complete cases and missing-data encodings, H1 failed only where

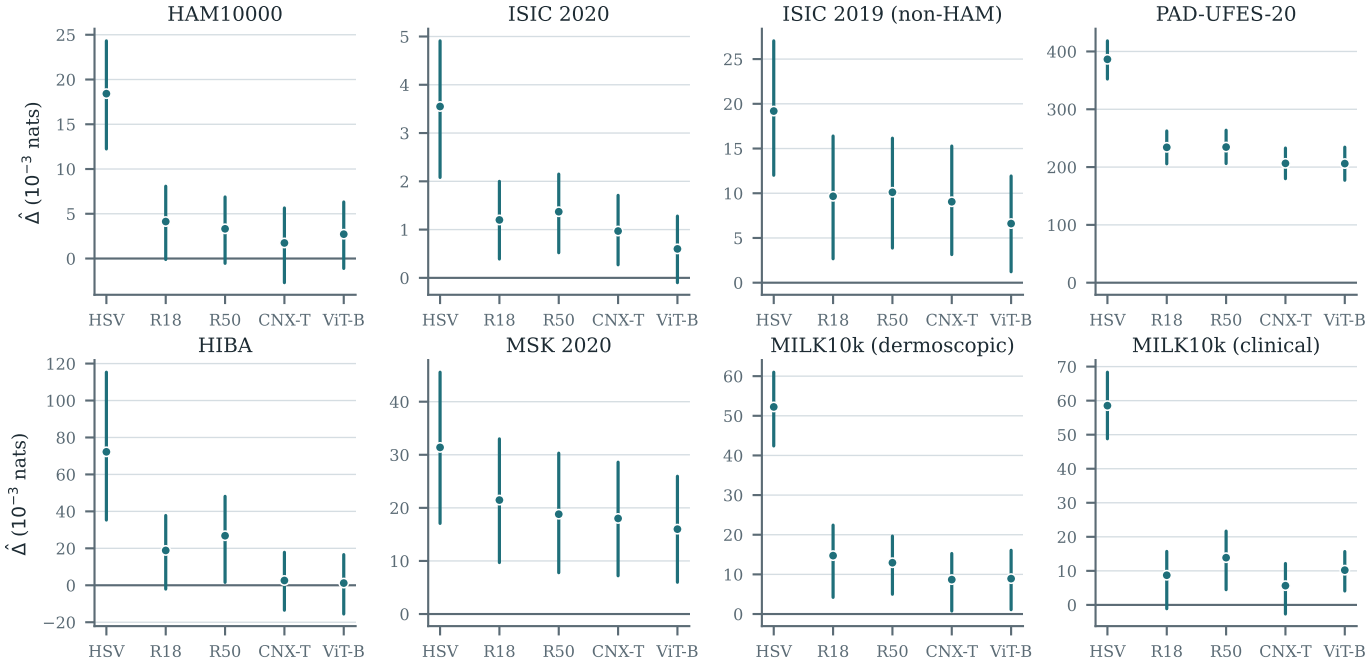


Fig. 1. Added value of patient metadata $\hat{\Delta}$ (out-of-fold log-loss reduction, 10^{-3} nats) along the ladder for all eight image sets, with 95% grouped-bootstrap CIs (fold assignment 0, all images). Panels have different y-axis scales: PAD-UFES-20's values are about two orders of magnitude larger. HSV: colour histogram; R18/R50: ResNet-18/50; CNX-T: ConvNeXt-T; ViT-B: ViT-B/16.

TABLE V
Pre-registered replication in two waves: added value of all metadata $\hat{\Delta}$ (10^{-3} nats, fold assignment 0) along the ladder, and Holm-adjusted one-sided p-values for H1 ($\Delta_{\text{HSV}} > \Delta_{\text{ConvNeXt}}$) and H2 ($\Delta_{\text{ResNet-18}} > \Delta_{\text{ConvNeXt}}$), grouped paired bootstrap. Holm families as registered: 6 tests (wave 1) and 8 tests (wave 2). The two MILK10k rows show the same lesions and are not independent.

Wave	Dataset	HSV	ResNet-18	ResNet-50	ConvNeXt-T	ViT-B/16	H1 p_{Holm}	H2 p_{Holm}
1	ISIC 2020	3.6	1.2	1.4	1.0	0.6	0.003	0.092
1	ISIC 2019 (non-HAM)	19.2	9.7	10.1	9.1	6.6	0.003	0.241
1	PAD-UFES-20	386.4	234.2	234.6	206.4	206.1	0.003	0.003
2	HIBA	72.2	18.9	26.8	2.6	1.2	0.004	0.004
2	MSK 2020	31.4	21.5	18.8	18.0	16.0	0.004	0.086
2	MILK10k (derm.)	52.2	14.7	12.9	8.7	8.9	0.004	0.004
2	MILK10k (clin.)	58.5	8.7	13.9	5.6	10.2	0.004	0.004

the metadata carried almost no information (MILK10k, melanoma vs. rest: 1.8×10^{-3} nats), whereas H2 depended on the estimator, the label, the fold assignment and the weighting (Appendix H). H1 also held under full-feature fusion in 8 and under gradient-boosted fusion in 8 of 8 image sets (Appendix G). At ConvNeXt-T the remaining added value (10^{-3} nats, 95% CI) is: ISIC 2020 1.0 [0.3, 1.7]; ISIC 2019 (non-HAM) 9.1 [3.1, 15.3]; PAD-UFES-20 206.4 [179.9, 232.9]; HIBA 2.6 [−13.5, 17.9]; MSK 2020 18.0 [7.2, 28.6]; MILK10k (derm.) 8.7 [0.8, 15.3]; MILK10k (clin.) 5.6 [−2.7, 12.2].

Mechanism test 1 (PAD-UFES-20, wave 1). We predicted that symptoms, being invisible in photographs, would keep their value while demographics attenuated. From HSV to ConvNeXt-T, the demographic block retained 0.506 of its added value and the symptom block 0.474, so neither M1 nor M2 held. The premise was false: even within a diagnostic class, ConvNeXt-T features predict the symptoms with AUROC 0.64 to 0.76 (bleeding and elevation are visible).

D. Pre-registered modality and lesion-size tests

Mechanism test 2 (P1, MILK10k). If metadata substitute for image detail, clinical close-ups, which show less detail than dermoscopy, should leave more of the metadata's information non-redundant. On the same 5,117 lesions the opposite held: $\rho = 0.069$ for clinical vs. 0.105 for dermoscopic images at ConvNeXt-T, a paired difference of −0.036 [−0.071, −0.004]. P1 is therefore rejected. A plausible reason is that its premise again failed in our setting: the two image types gave almost identical image-only AUROC (0.842 vs. 0.845), so the clinical representation was not weaker, consistent with a

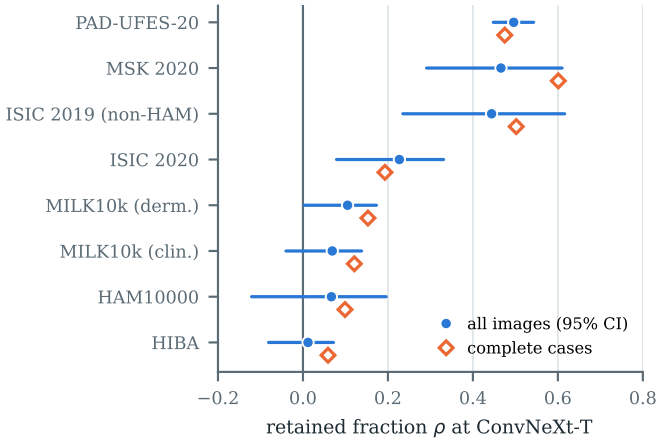


Fig. 2. Retained fraction ρ of the metadata's standalone information left non-redundant by ConvNeXt-T, per image set. Circles: all images with 95% grouped-bootstrap CI (Table VI); diamonds: complete cases (Table VII).

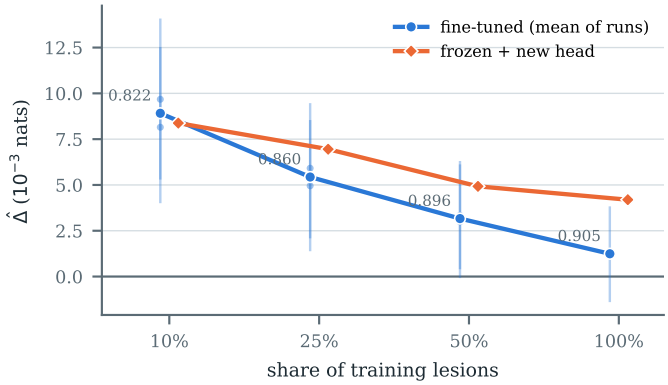


Fig. 3. Exploratory dose–response on HAM10000 (ResNet-18): added value of metadata vs. the share of training lesions available to the image model. All runs on complete cases. Blue: fine-tuned end to end; line through the mean of runs (two subsamples below 100%), small dots and bars are individual runs with 95% CIs, numbers are mean image AUROC. Orange: frozen ImageNet features with a retrained head (mean of two subsamples).

reader study that found the two modalities complementary rather than ranked [11].

P2 (MSK 2020). Measured lesion diameter added 8.6×10^{-3} nats [1.3, 16.4] to the colour-histogram model but only 1.8×10^{-3} [−1.7, 5.1] to ConvNeXt-T, a retained share of 0.21 (registered threshold < 0.5). P2 was registered as exploratory because MSK 2020 is small; the result is consistent with the finding below that deep image scores track lesion size.

E. How much information remains?

Normalising $\hat{\Delta}$ by the metadata's standalone information $\hat{I}_A$ gives the retained fraction ρ (Table VI, Fig. 2). It falls steeply from colour histograms to the deep representations everywhere, but its level at ConvNeXt-T varies from 0.01 (HIBA) to 0.50 (PAD-UFES-20). Dermoscopic datasets lie at both ends of this range, so the floor is a property of the dataset, its label definition and its metadata, not of the imaging modality. The label matters directly: on ISIC 2019, ρ is 0.48 for melanoma but 0.20 [0.14, 0.25] for malignant vs. benign (corrected estimator). Where added value is small, weighting and encoding matter too: HIBA's value at ConvNeXt-T is 3.8 per image but −0.8 per patient, and 12.7 without missing-value indicators (10^{-3} nats). The fusion estimator matters as well: whether any value remains at ConvNeXt-T changes with the estimator in 5 of 8 image sets, and gradient-boosted fusion recovers more value than score fusion in 5 (by 4.5–7.4$\times 10^{-3}$ nats outside PAD-UFES-20; Table XXIV), so the levels reported here are conservative.

F. Complete-case sensitivity analysis

Restricting every dataset to images with all metadata recorded (Table VII) removes label-informative missingness. H1 holds (uncorrected) in 8 of 8 image sets and H2 in 5 (not in ISIC 2019 (non-HAM), PAD-UFES-20, MSK 2020). On PAD-UFES-20, the standalone information in the metadata falls from 416×10^{-3} to 155×10^{-3} nats, so much of that dataset's apparent metadata signal reflects how missing values arise; its complete cases (1,089 images, mostly biopsied) are, however, a different population. Dropping the missing-value indicators instead, with median or multiple imputation fitted in the training fold, halved PAD-UFES-20's added value at ConvNeXt-T (206.9 to 112.2 and 90.7×10^{-3} nats) but raised HIBA's (3.8 to 12.7); H1's direction held under every encoding. A second stage given only the image score and missing-value indicators recovered 77% of PAD-UFES-20's added value at ConvNeXt-T and 25% of ISIC 2019's (all images; Table XXIV). Complete-case retained fractions at ConvNeXt-T range from 0.06 to 0.60.

G. What makes a representation "strong"?

The ladder orders representations by image-only performance but changes architecture, size, and pretraining at once. Four exploratory manipulations separate these factors; their CIs are not corrected for multiplicity.

Fine-tuning. ResNet-18 and ConvNeXt-T were fine-tuned end to end with the same grouped folds and a fixed schedule (AdamW, learning rate 10^{-4}, six epochs, no early stopping on test folds). On HAM10000, fine-tuning ResNet-18 raised image AUROC to 0.92 and lowered added value from 4.1 to 0.3×10^{-3} nats [−3.0, 3.5], a paired drop of 3.8×10^{-3} [1.7, 5.9] (Table VIII). For ConvNeXt-T, whose frozen added value was already small, the drop was not detectable [−1.1, 2.8]. On HIBA and PAD-UFES-20 the drop was significant for both backbones, although on PAD-UFES-20 most of the value remained (ConvNeXt-T: 206.4 to 123.9×10^{-3} nats). On ISIC 2019 and MILK10k (dermoscopic; ResNet-18 only), fine-tuning raised image AUROC (0.802 to 0.836; 0.815 to 0.869) but added value fell only slightly (9.7 to 7.1; 14.7 to 13.7×10^{-3} nats),

TABLE VI

Exploratory: retained fraction $\rho_k = \hat{\Delta}_k/\hat{I}_A$ of the metadata's standalone information that is not redundant with image representation k (all images, pre-registered encoding). $\hat{I}_A$: out-of-fold log-loss reduction of a metadata-only model over prevalence (10^{-3} nats). 95% grouped-bootstrap CIs, fold assignment 0. ρ_k can exceed 1 under synergy and fall below 0 in finite samples.

Dataset	$\hat{I}_A$	HSV	ResNet-18	ConvNeXt-T	ViT-B/16
HAM10000	25.9	0.71 [0.60, 0.80]	0.16 [0.00, 0.27]	0.07 [−0.12, 0.20]	0.10 [−0.03, 0.21]
ISIC 2020	4.3	0.83 [0.72, 0.95]	0.28 [0.15, 0.38]	0.23 [0.08, 0.33]	0.14 [−0.04, 0.25]
ISIC 2019 (non-HAM)	20.4	0.94 [0.84, 1.04]	0.47 [0.25, 0.66]	0.44 [0.23, 0.62]	0.33 [0.12, 0.48]
PAD-UFES-20	416.3	0.93 [0.90, 0.95]	0.56 [0.51, 0.61]	0.50 [0.45, 0.54]	0.49 [0.45, 0.54]
HIBA	217.4	0.33 [0.18, 0.51]	0.09 [−0.01, 0.16]	0.01 [−0.08, 0.07]	0.01 [−0.09, 0.07]
MSK 2020	38.7	0.81 [0.68, 0.92]	0.56 [0.39, 0.69]	0.47 [0.29, 0.61]	0.41 [0.22, 0.57]
MILK10k (derm.)	82.3	0.63 [0.58, 0.68]	0.18 [0.06, 0.26]	0.10 [0.00, 0.17]	0.11 [0.00, 0.18]
MILK10k (clin.)	82.3	0.71 [0.66, 0.76]	0.11 [−0.01, 0.18]	0.07 [−0.04, 0.14]	0.12 [0.05, 0.17]

TABLE VII

Complete-case sensitivity analysis: only images with every metadata field recorded (fold assignment 0; one-sided paired grouped-bootstrap p-values, uncorrected). $\hat{I}_A$ and $\hat{\Delta}$ in 10^{-3} nats; ρ at ConvNeXt-T.

Dataset	Kept (dropped)	$\hat{I}_A$	$\hat{\Delta}_{\text{HSV}}$	$\hat{\Delta}_{\text{ConvNeXt}}$	H1 p	H2 p	ρ
HAM10000	9,761 (254)	26.3	18.1	2.6	≤0.001	0.018	0.10
ISIC 2020	32,531 (595)	4.1	3.4	0.8	≤0.001	≤0.001	0.19
ISIC 2019 (non-HAM)	14,156 (1,160)	12.2	13.8	6.1	0.002	0.162	0.50
PAD-UFES-20	1,089 (1,209)	155.4	152.0	73.8	≤0.001	0.520	0.47
HIBA	1,085 (139)	241.0	75.2	14.1	≤0.001	≤0.001	0.06
MSK 2020	1,157 (17)	36.2	33.6	21.8	0.009	0.073	0.60
MILK10k (derm.)	5,086 (31)	80.5	50.7	12.3	≤0.001	≤0.001	0.15
MILK10k (clin.)	5,086 (31)	80.5	57.6	9.8	≤0.001	≤0.001	0.12

TABLE VIII

Exploratory: frozen vs. end-to-end fine-tuned image models (same grouped folds, fold assignment 0, all images; one training run per fold). $\hat{\Delta}$ and the paired difference (frozen − fine-tuned) in 10^{-3} nats with 95% grouped-bootstrap CI. PAD-UFES-20 includes its label-informative missingness.

Dataset	Backbone	AUROC frz./FT	$\hat{\Delta}$ frz./FT	Difference
HAM10000	ResNet-18	0.854 / 0.917	4.1 / 0.3	3.8 [1.7, 5.9]
HAM10000	ConvNeXt-T	0.887 / 0.930	1.7 / 0.8	0.9 [−1.1, 2.8]
HIBA	ResNet-18	0.908 / 0.940	18.9 / 1.4	17.5 [6.1, 29.8]
HIBA	ConvNeXt-T	0.928 / 0.951	2.6 / −7.2	9.8 [2.6, 17.4]
PAD-UFES-20	ResNet-18	0.845 / 0.899	234.2 / 172.7	61.5 [43.8, 79.5]
PAD-UFES-20	ConvNeXt-T	0.874 / 0.927	206.4 / 123.9	82.5 [65.5, 99.0]
ISIC 2019 (non-HAM)	ResNet-18	0.802 / 0.836	9.7 / 7.1	2.6 [−0.2, 5.7]
MILK10k (derm.)	ResNet-18	0.815 / 0.869	14.7 / 13.7	1.0 [−2.4, 4.5]

and neither paired difference excluded zero ([−0.2, 5.7]; [−2.4, 4.5]). With two further training seeds on HIBA and PAD-UFES-20, the drop held in 12 of 12 seed–backbone–dataset runs, and added value varied across seeds by at most 6.7×10^{-3} nats.

Training-set size. On HAM10000's complete cases, ResNet-18 was fine-tuned on 10%, 25%, 50%, and 100% of each training fold's lesions, with two independent subsamples below 100% (7 runs). Mean added value fell from 8.9 to 5.4, 3.2, and 1.2×10^{-3} nats (100%: 95% CI [−1.3, 3.8]) as image AUROC rose from 0.822 to 0.905 (Fig. 3); across runs, the Spearman correlation between training fraction and added value was −0.97. Between 50% and 100%, image AUROC changed little (0.896 vs. 0.905) while added value kept falling, although the intervals of adjacent fractions overlap; image AUROC is thus an incomplete summary of representation strength. Retraining only the head of frozen features on the same fractions moved added value in the same direction but less (8.4 to 4.2 on HAM10000; 10.2 to 7.0 on ISIC 2019).

Backbone size. Within families (ResNet-18/34/50/101; ConvNeXt-T/S/B), frozen and on complete cases of 3 datasets, none of 6 smallest-vs-largest comparisons of added value was significant, and image AUROC changed by only −0.016 to +0.009.

A larger pretrained model. A frozen ViT-L/16 (304M parameters, weakly supervised SWAG pretraining [28]) changed image AUROC relative to ConvNeXt-T by −0.005 to +0.020 across 8 image sets; added value was slightly lower on ISIC 2020, ISIC 2019 (non-HAM) and not higher anywhere.

Together, these results tie attenuation to how well the image model solves the task, not to the number of parameters applied to frozen features.

H. Erasing what the image encodes about age

Proposition 1 predicts that if a variable adds little because the representation already encodes it, removing that encoding should restore its value. We tested this with linear concept erasure [16], registered in advance with a confirmatory family of 4 tests (age on ISIC 2020, HIBA and MSK 2020; anatomical site on HIBA; Holm) on datasets not used in earlier erasure experiments. In each training fold, the closed-form projection that makes age linearly undecodable was fitted to the image features and applied to all images. The primary contrast compares age's

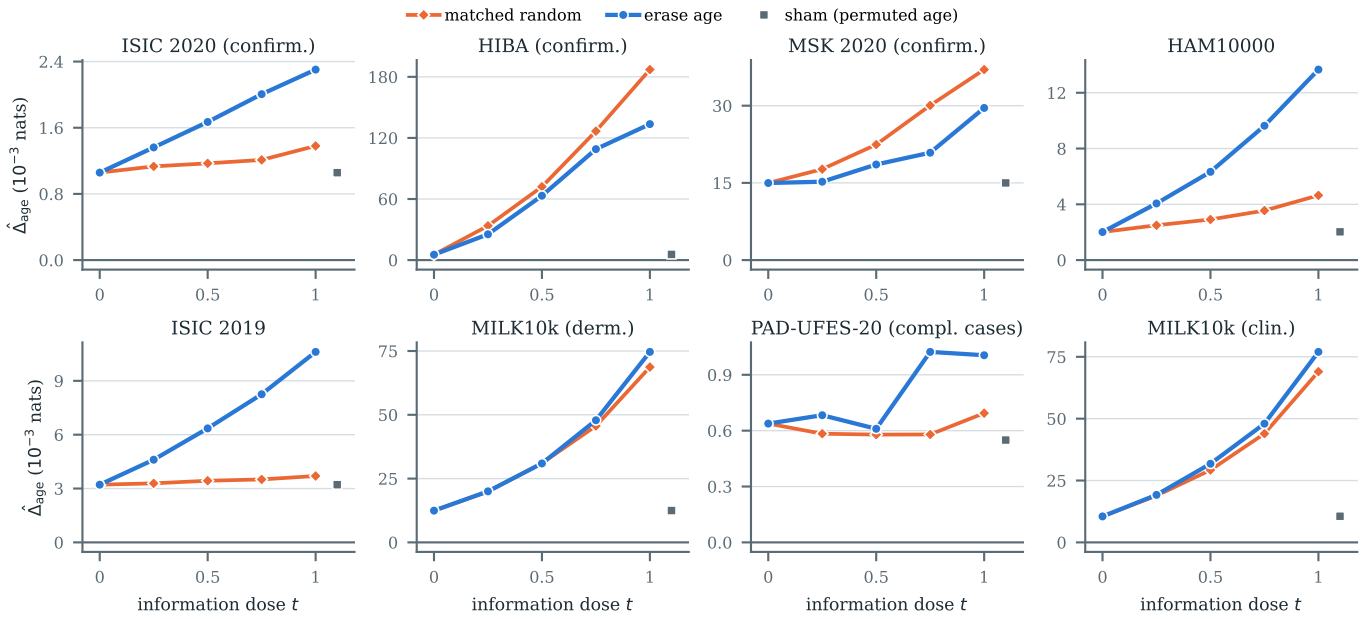


Fig. 4. Erasing the linear trace of age (E50, ConvNeXt-T, fold assignment 0, strictly cross-fitted): added value of age as the erased coordinates are mixed with independent noise (information dose t; $t = 0$: unchanged features; $t = 1$: coordinates replaced by noise), for the age eraser and for random subspaces whose removal costs as much image performance (mean of five). Squares: shams, erasers fitted to permuted age (mean of five). The first three panels are the registered confirmatory datasets; HIBA and MSK 2020 fail for the reasons in Section V-H. Panels have their own vertical scales.

TABLE IX

Erasing the linear trace of age (E50, ConvNeXt-T, fold assignment 0; 10^{-3} nats). R^2: age decodability in held-out images before and after erasure (within-test-fold ridge probe). $\hat{\Delta}_{age}$: added value of age with unchanged features, after erasure, and after removing performance-matched random subspaces (mean of 5; per-fold ranks chosen by inner cross-validation on training data only). Contrasts are paired differences with 95% grouped-bootstrap CIs. Sham: eraser fitted to permuted age (mean of 5); †95% CI above 0. Matching check: image-only log-loss, erasure minus matched control (negative: the control removed more image information, so the contrast is conservative). SD ratio: test/training standard deviation of the erased coordinate (overfit eraser if far from 1). Confirm.: registered confirmatory family (Holm, 4 tests); Replic.: datasets used in earlier erasure experiments; Correl.: shares lesions with MILK10k (derm.). ‡Complete cases.

				$\hat{\Delta}_{age}$						
Image set	Role	Matched rank	R^2 (t=0 → 1)	t=0	Erased	Matched	Erased − matched	Erased − sham	Matching check	SD ratio
ISIC 2020	Confirm.	32–128	0.23 → −0.03	1.06	2.27	1.27	0.99 [0.69, 1.29]	1.21†	−0.37 [−1.19, 0.38]	0.99
HIBA	Confirm.	512	0.35 → −0.06	5.33	122.51	239.07	−116.56 [−160.01, −78.81]	116.93†	−125.62 [−157.63, −94.36]	1.80
MSK 2020	Confirm.	256–512	0.19 → 0.03	14.98	25.25	49.09	−23.84 [−30.07, −17.86]	10.25†	−38.39 [−50.72, −26.39]	2.21
HAM10000	Replic.	91–181	0.33 → 0.00	2.01	13.19	4.14	9.05 [6.48, 11.70]	11.16†	−11.43 [−15.76, −7.36]	1.01
ISIC 2019	Replic.	23–91	0.32 → 0.01	3.22	10.45	3.52	6.92 [4.76, 9.14]	7.23†	−3.99 [−7.38, −0.40]	1.01
MILK10k (derm.)	Replic.	512	0.28 → 0.02	12.44	69.02	59.50	9.53 [4.50, 14.20]	56.54†	−38.67 [−46.77, −29.79]	1.04
PAD-UFES-20‡	Replic.	1–11	0.09 → 0.00	0.64	0.88	0.67	0.22 [−2.63, 3.11]	0.33	−1.21 [−6.53, 4.01]	2.98
MILK10k (clin.)	Correl.	512	0.34 → 0.01	10.50	70.59	58.96	11.63 [6.67, 16.59]	60.02†	−34.88 [−43.28, −26.21]	1.05

added value after erasure with its mean over five random subspaces whose removal costs the image model as much performance, their rank chosen in each fold by inner cross-validation on training data only. Shams (erasers fitted to permuted age), an information dose that mixes the erased coordinates with noise, re-appending the erased coordinates, and a cross-fitted image-derived age estimate probe the mechanism further (Tables IX and XXI).

On ISIC 2020, erasure raised age's added value by 0.99 [0.69, 1.29]$\times 10^{-3}$ nats more than the matched control (Holm $p = 0.002$; Fig. 4, Table IX) and by 1.21 [0.86, 1.55] more than the shams; the contrast was 0.85–1.08 across three fold assignments and 1.49 [0.82, 2.20] with a full-feature fusion model (Table XXII). The replication datasets on which age is decodable gave the same result (ConvNeXt-T: HAM10000 9.05 [6.48, 11.70], ISIC 2019 6.92 [4.76, 9.14], MILK10k 9.53 [4.50, 14.20]; ResNet-18: 8.89 [6.37, 11.31], 6.22 [4.08, 8.37], 22.19 [17.02, 27.22]), so all 6 of 6 earlier exploratory claims survive the stricter control, although MILK10k's ConvNeXt-T estimate fell from 34.7 because its matched control now removes more image information. On ISIC 2020, HAM10000 and ISIC 2019, the contrast grew with the information dose and excluded zero at every dose; erasure raised age's added value by 0.71–0.97 nats per nat of image log-loss lost, against 0.03–0.10 for matched random subspaces; and

the image-derived age estimate recovered the rise within its CI (Table XXI). Re-appending the erased coordinates restored image performance fully but returned only 74–93% of the rise.

The effect was specific to the erased variable. Erasing anatomical site instead raised site's added value beyond its matched control on HAM10000 (4.26 [1.75, 6.86]) and MILK10k (3.83 [1.23, 6.31]), and on both datasets each eraser raised the other variable's value no more than its matched control did (age eraser, site's value: 0.39 [−0.97, 1.69] and −1.47 [−2.65, −0.25]; site eraser, age's value: 0.55 [−0.49, 1.61] and −1.15 [−2.48, 0.19]), satisfying the registered selectivity rule in both directions (Table XXIII). Erasure removes linearly decodable information only: on ISIC 2020, held-out linear decodability of age fell from $R^2 = 0.23$ to -0.03, but a registered one-hidden-layer probe still decoded age after erasure on all five large image sets (R^2 0.05–0.23), about as well as after matched random erasure (ISIC 2020: 0.23 before, 0.20 after erasure, 0.20 after random erasure). The redundancy identified here is therefore with the information that a linear read-out of the representation uses.

The other three confirmatory tests failed (age on HIBA, −116.6; MSK 2020, −23.8; site on HIBA, -54.1×10^{-3} nats; all $p = 1$), and the design explains why it cannot work there. These datasets have 871–980 training images for 768 features, and the fitted eraser does not generalise: its coordinate is 1.8–3.1 times as variable in test as in training images (larger datasets: 0.99–1.08), so random subspaces matched on training data do not match the eraser's damage on test images. On HIBA and MSK 2020 the label also depends strongly on age (age alone predicts it with AUROC 0.84 and 0.72, against 0.67 on ISIC 2020), so erasing age removed diagnostic signal itself (HIBA image AUROC 0.93 to 0.65), the known failure of probe-guided removal when concept and label are correlated [18]. On PAD-UFES-20, age was barely decodable and erasure had no effect (0.22 [−2.63, 3.11]).

Marginal erasure removes all of the image's training covariance with age, including covariance that arises because the label depends on age. That erasing age cost image performance (Table IX) therefore shows that the image model's linear read-out depends on directions that covary with age, not that it uses age as such. An exploratory class-conditional eraser, defined after the confirmatory results, removes only the age trace not explained by the label; it gave the same positive, well-matched result on all five large image sets (0.86–14.05×10^{-3} nats) at a 15–77% smaller cost in image log-loss (Table XXII). These results show the substitution term of Proposition 1 at work for age, and for site on two datasets, for linearly decodable information; they do not show that training learns age in order to diagnose.

I. Where do metadata help?

Pre-registered before analysis, we built for each training fold a nearest-neighbour graph of the image representation and asked whether per-image benefit is larger where a lesion's neighbours have discordant training labels (H1), where the image model is uncertain (H2), and whether graph statistics improve held-out prediction of benefit beyond the fitted models (H4; Appendix I). At ConvNeXt-T, H2 held in 7 of 7 dataset units, as the bound $u(s) \leq H(Y \mid S = s)$ requires; H1 held in 4 (registered threshold: 5) and H4 in 0. Post hoc, the fusion model's label-free expected gain K ranked per-image benefit best in 6 of 8 image sets, and the quarter of images with the largest K carried 70%–105% of the total benefit (HIBA excluded: total near zero), although K overstated the level of benefit, by up to 9-fold. On ConvNeXt-T, benefit showed no local organisation on the graph beyond label and image uncertainty (normalised Dirichlet energy at most 1% below a stratified permutation null, significant in 2 of 8 sets; colour histograms: 6). In the decomposition $\Delta = H\eta$, falling image uncertainty accounted for only 0.09–0.33 of the log-attenuation from colour histograms to ConvNeXt-T on seven sets (bootstrap upper bounds $\leq$ 0.58), against 0.61 [0.55, 0.66] on PAD-UFES-20 (exploratory): strong representations attenuate metadata mainly by leaving uncertainty that metadata no longer resolve, not by resolving more cases.

VI. Discussion and Limitations

What the results mean. On every independent dataset, metadata add clearly more to a colour-histogram model than to frozen deep representations, and better task-specific image models (fine-tuned, or trained on more data) leave less for the metadata to add, although on ISIC 2019 and MILK10k the drop after fine-tuning was small and not significant. Attenuation between frozen deep representations is smaller and inconsistent (H2: 3 of 6), as expected if those representations differ little in task performance. How much value remains at a strong representation depends on the dataset and its label definition, from little (HAM10000; HIBA, depending on weighting and missing-value encoding) to about half (PAD-UFES-20, MSK 2020, ISIC 2019 for melanoma), and on the fusion estimator, but not on imaging modality: the pre-registered prediction that clinical images leave more value than dermoscopy of the same lesions failed. Concept erasure shows the redundancy at work for age: removing the linear trace of age raised age's added value by 0.71–0.97 nats per nat of image log-loss lost, against 0.03–0.10 for matched random removal, on a pre-registered held-out dataset and two replications. The frequently reported benefit of metadata [1], [2] is therefore partly a statement about weak image representations, and the size of the part that is not differs severalfold between datasets.

Which variables keep their value? In an exploratory analysis over 95 dataset–representation–variable combi-

nations (of 160; the rest had no significant standalone value), variables that the image predicts well within a diagnostic class kept a smaller share of their value (Spearman -0.75; partial correlation adjusting for standalone value and image AUROC -0.59, dataset-clustered 95% CI [-0.72, -0.40]; between -0.41 and -0.75 within each representation). This is the pattern Proposition 1 predicts, but it is correlational, and it holds only among variables with demonstrable standalone value: there it survives a mixed model with dataset effects (coefficient -0.49 [-0.67, -0.31]), whereas over all 160 combinations it weakens (partial -0.28; mixed -0.10 [-0.25, 0.05]). A transparent rule fitted to these quantities did not predict a variable's added value in held-out datasets better than predicting zero (mean absolute error 0.0101 vs. 0.0092 nats); the simple product of standalone value and a typical retained fraction did best (0.0082). As a yes/no decision (collect if the predicted value is at least 10^{-3} nats), that simple product was right in 0.80 of cases (precision 0.81, recall 0.83; Appendix E). With six independent datasets, a reliable quantitative rule is beyond reach.

Under dataset shift. When the image and fusion models were fitted on one dataset and applied to another (harmonised age, sex, and site; 26 directed pairs), metadata improved log-loss significantly in 14 pairs and worsened it in 7, from -0.147 to 0.063 nats. The largest harms occurred between datasets with very different prevalences and label definitions, so these results describe the joint shift rather than a property of metadata. Even within one dataset, added value learned on some sources did not transfer: holding out one HAM10000 source at a time (all images, strictly cross-fitted, ConvNeXt-T, 10^{-3} nats): rosendahl -6.5 [-16.8, 3.6]; vidir_modern -14.4 [-32.1, 0.9]; vidir_molemax 7.6 [6.1, 9.1]; vienna_dias 18.0 [3.2, 30.6]; ISIC 2019, BCN to MSK -10.5. Deciding from the source datasets alone whether to use metadata did not help (leave one dataset out within a label family, 7 targets; Table XX): mean regret was 3.6 for always using metadata, 3.9 for a source-validated rule and 3.6 for the simple product rule above, which recommended metadata on 7 of 7 targets, including PAD-UFES-20, where they hurt (-19.61 [-36.42, -2.41]$\times 10^{-3}$ nats).

Practical implication. Metadata fusion should be evaluated against the strongest available image model, ideally one fine-tuned on the target data, and the retained fraction ρ reported alongside the gain, under more than one fusion estimator, so that a benefit that merely compensates for a weak backbone is not mistaken for complementary information. Because benefit is concentrated, the fusion model's expected gain can rank the lesions whose metadata matter most, although it overstates the size of the gain.

Limitations. (i) Fine-tuning was examined on five datasets (two with ResNet-18 only) with one training run per fold (three seeds on HIBA and PAD-UFES-20, where the drop held for every seed) and a fixed schedule; the drop in added value was significant on three, and fine-tuned scores were not re-estimated with the strictly cross-fitted second stage. (ii) $\hat{\Delta}$ estimates the reduction achievable by a logistic second stage on a one-dimensional score, not $I(Y; A \mid S)$ itself; full-feature and boosted fusion recover different amounts, so per-dataset levels at strong representations depend on the estimator, although H1 does not (Table XXIV). (iii) The frozen ladder is ordered by image-only performance and is not a controlled intervention; Section V-G addresses this only in part. (iv) Both pre-registered mechanism tests failed because their premises (invisible symptoms; a weaker clinical-image representation) did not hold; the mechanism is shown by pre-registered intervention for age on one confirmatory dataset and three replications, and for site on two, and only for linearly decodable information; the erasure design needs many more training images than features and a label that does not depend strongly on the erased variable, and was uninformative on HIBA, MSK 2020 and PAD-UFES-20. (v) Images come from public archives and mirrors; official metadata were used and counts were checked against the published descriptions. MSK 2020 has no lesion identifier and is grouped per image (an embedding check found no near-duplicate images within it); ISIC 2019 images without one are grouped individually. (vi) Missingness is label-informative in every dataset, severely so in PAD-UFES-20; complete-case analysis removes this leak but analyses a selected subsample, and some exploratory PAD-UFES-20 results (Table VIII) use all images. (vii) All analyses are retrospective; nothing here is a clinical claim, and no causal effect of metadata is estimated. (viii) Bootstrap CIs hold the fitted models fixed and omit training variability [23]: across fold assignments, added value at ConvNeXt-T varied by 22.2×10^{-3} nats on PAD-UFES-20 (CI half-width 27.1) and HIBA's took the values 3.8, 10.0, 9.1. (ix) Saliency analyses (Appendix J) are evidence of spatial correspondence, not of causal reliance [29].

VII. Conclusion

The value of patient metadata to a skin lesion classifier is not fixed: it depends on what the image model already extracts. Proposition 1 makes this precise. Pre-registered experiments on seven datasets show that metadata add far less to deep representations than to colour histograms, a result that survives a strictly cross-fitted estimator, alternative labels, complete cases and missing-data encoding. Interventions show what drives it: fine-tuning and more training data tend to shrink the gain, more parameters on frozen features do not, and removing the linear trace of age from a representation raises age's value beyond matched controls, in a pre-registered test on a held-out dataset and in three replications. Attenuation among frozen deep representations did not replicate consistently, the share of metadata information left non-redundant ranged severalfold across datasets and with the label definition, imaging modality did not explain that range, and the

remaining benefit concentrated in lesions the image model found uncertain. Claims that metadata improve a classifier should therefore be made against strong, task-adapted image baselines and accompanied by the retained fraction.

Appendix A
Index of experiments

Table X lists every experiment of this study and where it is reported. Earlier experiments in the same project (E01–E08: an Alzheimer's MRI study; E11: industrial anomaly detection) address unrelated questions and are not part of this paper.

Appendix B
Simulation of the attenuation signature

In a simulation with a known answer (E09), a latent image variable generated the label, and an auxiliary variable acted either through the image, directly on the label, both, or not at all. A ladder of image models was created by adding decreasing noise to the image representation. Added value fell to zero when the auxiliary variable acted through the image, as its recoverability from the image rose, and rose when it acted directly (Table XI); the latter is the non-collapsibility of logistic regression [13] and corresponds to $R(S) < 0$ in Proposition 1. The mixed case fell to a non-zero floor.

Appendix C
Further manipulations of the image representation

Tables XII–XIV give the full results summarised in Section V-G. Table XV reports a resolution ladder (E19): with the architecture fixed, images were downsampled before embedding. The prediction that added value falls monotonically with resolution held in 3 of the 7 image sets analysed (PAD-UFES-20, MILK10k (derm.), MILK10k (clin.); the PAD-UFES-20 analysis uses all images and therefore includes its missingness leak), and image AUROC hardly changed (on HAM10000 it is 0.88 even at 16×16 pixels), so resolution is a weak handle on representation strength for these tasks.

Appendix D
Per-variable floors

Single-variable added values (Table XVI) show that every variable with detectable value at the colour-histogram rung attenuates on HAM10000 and ISIC 2020 (on ISIC 2020, sex and site carry no detectable value at all), whereas on ISIC 2019 anatomical site keeps most of its value. This is consistent with ISIC 2019's large non-redundant share (Table VI) and with BCN20000's deliberate inclusion of hard-to-diagnose sites [5].

Appendix E
Exploratory analyses

Recoverability (Fig. 5). The negative association reported in Section VI does not depend on the colour histogram: restricted to the deep representations, the within-dataset Spearman correlation is negative in 8 of 8 image sets (from -0.40 to -0.95; as few as 4 points in some sets). The rule excluding rows with non-significant standalone value was fixed in the analysis code before the run but was not stated in the dated registry.

Permutation control (Table XVII). Permuting whole metadata records between grouping units destroys their link to the image and label but keeps their joint distribution; permuted metadata never matched the real metadata's added value and always had negative added value on average, as expected from fitting extra parameters to noise.

Collection rule (Table XVIII). A ridge model on six cheap indicators was worse than predicting zero; the product of standalone value and the median retained share of the training datasets was the best predictor and made a useful collect/skip decision within the datasets used to fit it; under dataset shift it recommended metadata for every held-out dataset (Table XX).

Dataset shift (Table XIX). Under shift, metadata fitted on the source frequently harmed the target, most strongly between datasets with very different prevalences. Within HAM10000, holding out one image source at a time (10^{-3} nats, 95% CI): rosendahl -5.9 [-13.9, 2.0]; vidir_modern 6.3 [1.8, 11.2]; vidir_molemax 7.4 [6.2, 8.6]; vienna_dias 16.2 [5.2, 28.0].

Appendix F
Concept erasure: details

Erasers are fitted on each outer training fold's features and the attribute only (never diagnostic labels, except for the class-conditional variant, which uses training-fold labels to residualise the attribute) and applied to every image. With whitening W of the training features and U the left singular vectors of $W\Sigma_{XZ}$, the erased coordinates are $c = (x - \mu)WU$ and the erased features $x - cU^{\top}W^{+}$. The matched rank is the smallest value on $\{1, 2, 4, \ldots, 512\}$, refined at $\sqrt{2}$ steps, whose inner cross-validated image log-loss on the training fold is at least that of the eraser; random subspaces are drawn in the same whitened space. The training-fold ridge probe returns $R^2 \approx 0$ after any exact eraser by construction, so residual decodability is measured by a probe cross-validated within each held-out fold (Table IX). A Ledoit–Wolf-whitened variant of the eraser (registered as secondary) gave the same contrasts on the large datasets. On HIBA, MSK 2020 and PAD-UFES-20 every targeted eraser's coordinate was overdispersed in test images (Table IX), and the matching check failed in one direction or the other for every eraser there except PAD-UFES-20's age eraser, which had no effect (Tables IX, XXII, XXIII).

TABLE X
All experiments of this study. Pre-reg.: hypotheses, tests and correction fixed before modelling. Exploratory: registered or described before running where stated, but not confirmatory.

ID	Question	Status	Reported in	Outcome
E09	Does the slope of added value along a ladder separate mediated from direct information?	Simulation	App. B	Yes: mediated falls, direct rises
E10	Does added value attenuate along the ladder on HAM10000?	Pre-reg.	Sec. V-A, Tab. II	3 of 4 predictions held; no non-zero floor
E10a	Which metadata fields leak the label?	Audit	Sec. IV	dx_type, biopsed excluded; image source is a shortcut
E10b	Is attenuation a source shortcut? Paired tests	Exploratory	Tab. IV	Holds with source adjustment
E10c, E20	Which variables attenuate?	Exploratory	Sec. V-A, Tab. XVI	All attenuate together; ISIC 2019 floor carried by site
E10d	Effect in AUROC and Brier units	Exploratory	Tab. III	ΔAUROC 0.043 $\rightarrow$ 0.003
E12, E12b	Does evidence lie on the lesion? (ResNet-50)	Exploratory	Tab. XXVIII	Yes; sign-convention pitfall found
E13 (wave 1)	Replication on ISIC 2020, ISIC 2019, PAD-UFES-20	Pre-reg.	Tab. V	H1 3/3, H2 1/3; PAD mechanism M1–M3 failed
E13/E22 (wave 2)	Replication on HIBA, MSK 2020, MILK10k; P1, P2	Pre-reg.	Tab. V, Sec. V-D	H1 4/4, H2 3/4; P1 failed; P2 held
E14, E14b	Frozen vs. fine-tuned	Exploratory	Tab. VIII	Added value lower after fine-tuning in all 8 pairs; significant on HAM10000 (ResNet-18), HIBA, PAD-UFES-20
E31c, E31d	Fine-tuning on 10–100% of the data	Exploratory*	Fig. 3	Added value falls monotonically with training data
E15	Interpretability across backbones	Exploratory	App. J	Misses are small lesions; on-lesion evidence beyond centring in 10%–19% of maps (E48)
E16	Are PAD-UFES-20 symptoms visible?	Exploratory	Sec. V-C	Yes, partly, within class
E17	Retained fraction ρ	Exploratory	Tab. VI, Fig. 2	Dataset-specific floors
E18	Does a 304M-parameter ViT attenuate further?	Exploratory	Tab. XIV	No consistent change
E19	Does lower resolution restore added value?	Exploratory	Tab. XV	Weak manipulation; mixed
E21	Complete-case re-analysis	Exploratory	Tab. VII	H1 8/8, H2 5/8 (uncorrected)
E30	Does recoverability predict what is retained?	Exploratory*	Sec. VI, Fig. 5	Negative association within every representation
E31a, E31b	Backbone size; data size with frozen features	Exploratory*	Sec. V-G, Tabs. XII, XIII	Data and fine-tuning matter; size does not
E32	Permutation control	Exploratory*	Sec. V-B, Tab. XVII	Real metadata beat all permutations
E33	Cross-dataset shift	Exploratory*	Sec. VI, Tab. XIX	Mixed; confounded with prevalence
E34	A rule for which metadata to collect	Exploratory*	Sec. VI, Tab. XVIII	Fitted rule failed; simple baseline best
E40	Does a strictly cross-fitted second stage change the results?	Exploratory*	Sec. IV, App. H	Changes $\leq$0.42 CI half-widths; H1 unchanged; ISIC 2020 H2 now holds
E41	Alternative labels; complete cases, corrected	Exploratory*	Tab. XXV	H1 robust except MILK10k melanoma; ρ label-dependent
E42	Does graph ambiguity locate or predict benefit?	Pre-reg.†	Sec. V-I, Tab. XXVI	H1 4/7 (failed), H2 7/7, H4 0/7 (failed)
E44	Is benefit smooth on the graph?	Exploratory*	Sec. V-I, Tab. XXVI	Not beyond label and uncertainty (ConvNeXt-T)
E45, E45b	Does erasing age from the representation restore its value?	Exploratory*	Sec. V-H, Tab. IX	Yes where age is decodable; beyond matched random erasure in 6/8 pairs
E50	Does erasing age's linear trace raise its value beyond matched random and sham erasure?	Pre-reg.	Sec. V-H, Tabs. IX, XXI	Confirmatory 1/4 (ISIC 2020); 3 replications; uninformative where $n \approx d$
E50b, E50c	Fold assignments, ResNet-18; class-conditional erasure	Exploratory*	Tab. XXII	Contrast stable; class-conditional: same on 5 large sets (post hoc)
E51	Is erasure specific to the erased variable (site)?	Pre-reg.	Tab. XXIII	Selective for age and site on 2 datasets; HIBA failed
E52	Does a nonlinear probe still decode age after erasure?	Exploratory*	Sec. V-H, Tab. XXI	Yes: erasure is linear only
E53, E53b	Do results depend on the fusion estimator?	Exploratory*	Tabs. XXIV, XXII	H1 and the erasure contrast hold; levels at ConvNeXt-T do not
E54	Fine-tuning with further seeds	Exploratory*	Sec. V-G	Drop held for every seed (HIBA, PAD-UFES-20)
E55	Source-only rules for using metadata under shift	Exploratory*	Sec. VI, Tab. XX	None beat always using metadata
E46	E30 without its filter	Exploratory*	Sec. VI	Holds only for variables with standalone value
E47	Missing-data encoding; source-held-out evaluation	Exploratory*	Sec. V-F, Sec. VI	Encoding changes added value up to threefold; H1 direction unchanged
E48	Is the shift null valid for centred lesions?	Exploratory*	App. J, Tab. XXVII	No: a centred Gaussian passes in 88% of images
E49	Uncertainty for the $H\eta$ decomposition	Exploratory*	Sec. V-I	Image uncertainty explains 0.09–0.33 of attenuation (PAD-UFES-20: 0.61)

*Registered in a dated document before implementation.
†Hypotheses and decision rules fixed in a dated registry before implementation.

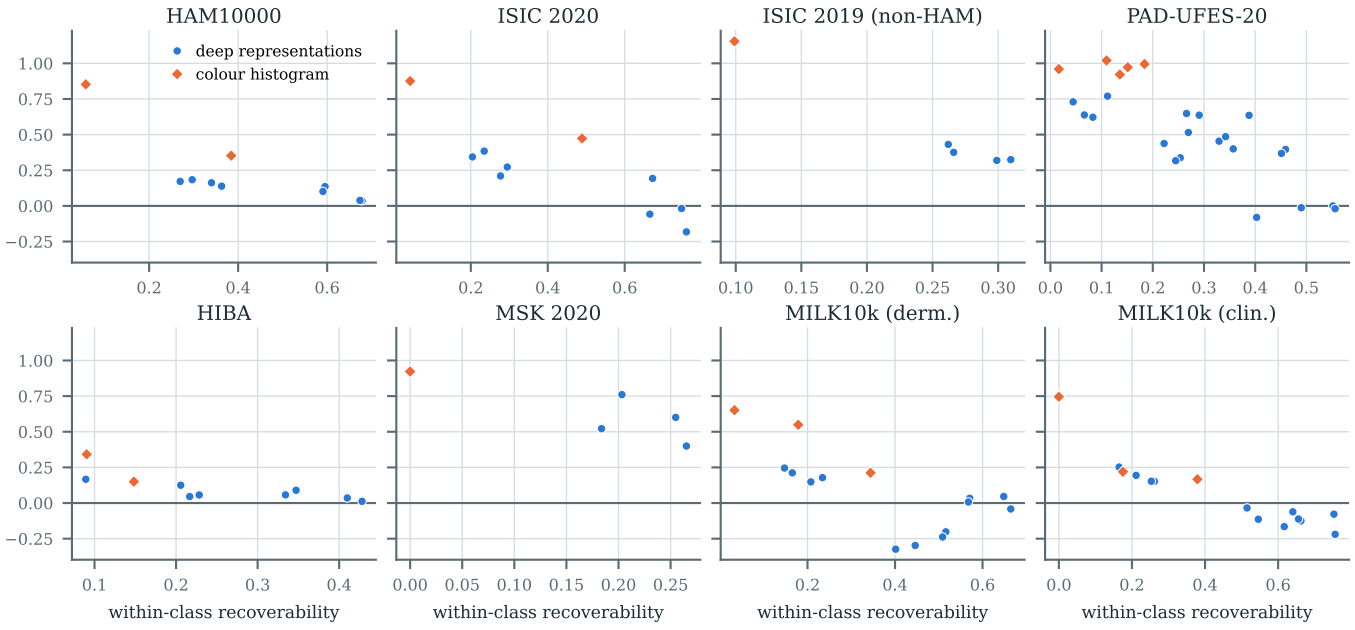


Fig. 5. Exploratory (E30): retained share of each variable's added value ($\hat{\Delta}_{\text{var}}/\hat{I}_{\text{var}}$) against its within-class recoverability from the image representation, for the 95 combinations where the variable's standalone value was significant. Orange: colour histogram; blue: deep representations.

TABLE XI
Simulation (E09; $n = 3{,}000$, 10 repetitions, 6 image-noise levels). Mean added value (10^{-3} nats) and recoverability (R^2) of the auxiliary variable from the image at the weakest and strongest image model.

	$\hat{\Delta}$		Recoverability	
Auxiliary effect	weakest	strongest	weakest	strongest
Through the image	2.4	−0.7	0.10	0.57
Direct	29.4	43.6	−0.02	−0.02
Both	8.2	1.6	0.11	0.58
None	−0.2	0.0	−0.02	−0.02

Appendix G
Fusion estimators

Table XXIV recomputes added value with the same strict cross-fitting under four second stages. H1 holds under each of the first three on every image set. The level of added value at ConvNeXt-T, and on several datasets whether it is distinguishable from zero, depends on the estimator; boosted fusion, which can model interactions between the image score and the metadata, usually recovers the most.

Appendix H
Robustness of the replication

Table XXV re-tests H1 and H2 under the conditions of the audit. Alternative labels: malignant vs. benign ({mel, bcc} on HAM10000; {MEL, BCC, SCC} on ISIC 2019) and melanoma vs. rest (PAD-UFES-20, HIBA, MSK 2020, MILK10k). H1 held under the alternative label in 5 of 6 independent datasets and on complete cases in 8 of 8 image sets; H2 held on complete cases in 6 of 8 (uncorrected).

Appendix I
Where metadata help: details

For each outer training fold, image features were standardised with training-fold statistics and L_2-normalised; each training image was linked to its 20 nearest training images from other grouping units (cosine distance, Gaussian weights with bandwidth equal to the median 20th-neighbour distance, union-symmetrised), and each test image to its 20 nearest training images, so no test label entered any statistic. Local label entropy uses the neighbours' training labels; graph-local benefit averages the neighbours' inner out-of-fold benefit computed within the training fold. Results were the same with 10 or 50 neighbours and with binary weights. H4 compared ridge regressions with and without graph statistics, both including K, image entropy, margin, feature norm and local density; both had low held-out R^2, because their training targets came from models fitted on less data, so the post hoc rank correlations in Table XXVI are the more informative comparison. Per-sample prediction of a feature's incremental value is established in dynamic feature selection [15]; these analyses ask only where the benefit lies.

Appendix J
Interpretability

Interpretability test. For a linear head w on global-average-pooled features $f = \frac{1}{|\Omega|}\sum_{u\in\Omega} F(u)$, the class

TABLE XII
Backbone size within families (E31a, exploratory; frozen, complete cases, fold assignment 0). Each cell: image AUROC / $\hat{\Delta}$ (10^{-3} nats). No smallest-vs-largest difference in $\hat{\Delta}$ was significant (paired grouped bootstrap).

Dataset	ResNet-18	ResNet-34	ResNet-50	ResNet-101	ConvNeXt-T	ConvNeXt-S	ConvNeXt-B
HAM10000	0.859 / 4.2	0.854 / 4.8	0.851 / 4.0	0.843 / 3.7	0.891 / 2.6	0.875 / 3.2	0.891 / 1.6
PAD-UFES-20	0.779 / 73.4	0.773 / 87.6	0.771 / 83.1	0.771 / 87.9	0.776 / 73.8	0.779 / 68.3	0.771 / 77.8
ISIC 2019 (non-HAM)	0.802 / 7.0	0.807 / 7.2	0.803 / 6.7	0.812 / 6.1	0.825 / 6.1	0.820 / 6.2	0.827 / 5.6

TABLE XIII
Training-set size with frozen features (E31b, exploratory; complete cases). The image head sees 10–100% of each training fold's groups; the metadata stage uses all folds. Each cell: image AUROC / $\hat{\Delta}$ (10^{-3} nats), mean of two subsamples below 100%.

Dataset	Features	10%	25%	50%	100%
HAM10000	ResNet-18	0.798 / 8.4	0.828 / 7.0	0.842 / 4.9	0.859 / 4.2
	ConvNeXt-T	0.810 / 7.4	0.852 / 3.5	0.877 / 3.8	0.891 / 2.6
ISIC 2019 (non-HAM)	ResNet-18	0.747 / 10.2	0.775 / 7.0	0.793 / 7.0	0.802 / 7.0
	ConvNeXt-T	0.767 / 9.0	0.797 / 6.8	0.811 / 5.8	0.825 / 6.1

TABLE XIV
A larger pretrained representation (E18, exploratory; frozen, all images, fold assignment 0): ConvNeXt-T vs. ViT-L/16 SWAG [28]. $\hat{\Delta}$ and the paired difference (ConvNeXt − ViT-L) in 10^{-3} nats.

	Image AUROC		$\hat{\Delta}$		
Dataset	CNX-T	ViT-L	CNX-T	ViT-L	Difference
HAM10000	0.887	0.896	1.7	3.4	−1.6 [−3.6, 0.4]
ISIC 2020	0.859	0.879	1.0	0.5	0.5 [0.2, 0.8]
ISIC 2019 (non-HAM)	0.823	0.828	9.1	6.7	2.4 [0.2, 4.7]
PAD-UFES-20	0.874	0.886	206.4	189.7	16.8 [−1.5, 34.7]
HIBA	0.928	0.936	2.6	−0.0	2.6 [−6.1, 10.8]
MSK 2020	0.727	0.730	18.0	18.7	−0.7 [−6.0, 4.6]
MILK10k (derm.)	0.845	0.840	8.7	8.2	0.5 [−1.9, 3.0]
MILK10k (clin.)	0.842	0.840	5.6	5.9	−0.2 [−2.0, 1.5]

activation map $c(u) = \sum_j w_j F_j(u)$ [30] decomposes the logit exactly. For a non-negative evidence map e and a binary mask m on a periodic grid Ω, the mass inside the mask after a shift by τ is

$$M(\tau) = \frac{\sum_u e(u)\, m(u-\tau)}{\sum_u e(u)} = \frac{(e \star m)(\tau)}{\|e\|_1}, \qquad (4)$$

a circular cross-correlation. Hence $\mathcal{F}^{-1}[\mathcal{F}(e)\overline{\mathcal{F}(m)}]$ yields $M(\tau)$ for all $|\Omega|$ shifts in $O(|\Omega| \log |\Omega|)$, and $|\{\tau : M(\tau) \geq M(0)\}|/|\Omega|$ is an exact p-value under the hypothesis that the mask's position is uniform over circular translations. Unlike pixel permutation, the shift null preserves the spatial autocorrelation of e.

For each backbone we use the exact class activation map of the out-of-fold head. For ConvNeXt-T, whose pooled features pass through a LayerNorm before the head, the map is exact up to a positive per-image factor: with per-image LayerNorm scale σ and affine weights γ, the logit equals a constant plus $\sigma^{-1}|\Omega|^{-1} \sum_u \sum_j \tilde{w}_j \big(F_j(u) - \bar{F}(u)\big)$ with $\tilde{w}_j = w_j \gamma_j / s_j$ and $\bar{F}(u)$ the channel mean. The overlap statistics are invariant to that factor, and the reconstructed logits matched the head's logits with correlation 1.000 for every backbone.

Mapping the predicted class's evidence produced an apparent link between errors and off-lesion attention ($p < 10^{-49}$; Table XXVIII). This is an artefact: for "not melanoma" predictions, the evidence for that class lies in the surrounding skin. With the melanoma-evidence map used for every image (Table XXIX), melanoma evidence is concentrated on the lesion in 84%–91% of images across the three backbones, a criterion that a centred map would also meet. Missed melanomas show no less on-lesion evidence; instead they are consistently smaller lesions ($p \leq 5 \times 10^{-12}$ for every backbone). Pixel permutation declares 82%–90% of maps significant, against 36%–56% under the circular-shift null, consistent with the inflation predicted by its disregard of spatial autocorrelation [20], [21].

The shift null is itself invalid here. It is exact only if, given the map, the mask's position is uniform over circular translations. Dermoscopic lesions are centred, so any centred map passes it. A fixed centred Gaussian, which carries no image information, was significant for 88% of images under the shift null and for 5% under a mask-swap null that compares each image's own mask with the masks of 199 images of similar lesion area (E48; Table XXVII). Under the swap null, 10%–19% of real maps were significant (shift null on the same images: 37%–57%): some lesion-specific evidence exists, but most of the shift-null significance reflects centring. Preserving spatial autocorrelation is necessary for a valid spatial null but not sufficient; the transformation must also leave the joint distribution of map and mask unchanged under the null.

TABLE XV

Resolution ladder (E19, exploratory): frozen ConvNeXt-T on images downsampled to $r \times r$ pixels and resized back. Image AUROC at the extremes; $\hat{\Delta}$ (10^{-3} nats) at each resolution; Spearman correlation of resolution with $\hat{\Delta}$ (prediction: -1). The manipulation is weak: image AUROC barely depends on resolution. All images (PAD-UFES-20 includes its missingness leak).

	Image AUROC		$\hat{\Delta}$ at resolution					
Dataset	16 px	224 px	16	32	64	112	224	Spearman
HAM10000	0.880	0.887	4.1	3.2	3.0	3.4	1.7	−0.7
ISIC 2020	0.850	0.859	1.3	1.2	0.8	0.9	1.0	−0.6
ISIC 2019 (non-HAM)	0.804	0.823	12.2	9.0	8.7	8.4	9.1	−0.4
PAD-UFES-20	0.849	0.874	246.5	228.7	224.9	212.6	206.4	−1.0
HIBA	0.924	0.928	5.7	6.9	9.4	10.3	2.6	0.0
MILK10k (derm.)	0.827	0.845	12.1	10.3	11.0	9.9	8.7	−0.9
MILK10k (clin.)	0.835	0.842	10.6	10.2	8.5	6.8	5.6	−1.0

TABLE XVI

Single-variable added value (E20, exploratory; 10^{-3} nats, 95% CI, all images). Retained: ConvNeXt-T / HSV, shown only where the HSV value is significant. PAD-UFES-20 is omitted: its per-variable results were driven by label-informative missingness (Section V-F).

Dataset	Variable	HSV	ConvNeXt-T	Retained
HAM10000	age	14.7 [9.9, 19.6]	2.0 [−1.0, 4.5]	0.13
	sex	2.5 [0.8, 4.2]	0.4 [0.0, 0.8]	0.14
	site	5.7 [2.2, 9.3]	1.1 [−1.6, 3.7]	0.19
ISIC 2020	age	3.3 [2.1, 4.4]	1.1 [0.5, 1.7]	0.32
	sex	0.2 [−0.0, 0.5]	0.1 [−0.1, 0.3]	–
	site	0.3 [−0.2, 0.8]	−0.1 [−0.4, 0.2]	–
ISIC 2019 (non-HAM)	age	11.2 [5.7, 16.5]	3.2 [0.6, 5.7]	0.28
	sex	−0.8 [−1.6, −0.2]	−0.2 [−0.8, 0.3]	–
	site	7.1 [2.2, 12.6]	5.4 [0.1, 10.7]	0.76

TABLE XVII

Permutation control (E32, exploratory; ConvNeXt-T, complete cases): $\hat{\Delta}$ (10^{-3} nats) with the real metadata and the mean and maximum over 50 permutations of whole metadata records between grouping units. Last column: % of records with an implausible combination (e.g. sex-specific site), real / permuted, all images.

Dataset	Real	Perm. mean	Perm. max	Implausible (%)
HAM10000	2.6	−2.3	0.0	0.2 / 0.6
ISIC 2020	0.8	−0.2	0.2	0.1 / 0.1
ISIC 2019 (non-HAM)	6.1	−2.2	0.2	0.3 / 2.2
PAD-UFES-20	73.8	−28.1	−8.0	0.2 / 2.3
HIBA	14.1	−8.2	3.0	0.2 / 1.5
MSK 2020	21.7	−3.2	2.3	0.0 / 0.3
MILK10k (derm.)	12.3	−1.0	0.2	0.2 / 2.6
MILK10k (clin.)	9.8	−1.1	0.3	0.2 / 2.6

TABLE XVIII

Predicting a variable's added value in a held-out dataset (E34, exploratory; leave-one-dataset-out, MILK10k held out as one; 160 dataset–representation–variable rows). MAE in 10^{-3} nats. Decision rule fixed in advance: collect if predicted $\hat{\Delta}_{\text{var}} \geq 10^{-3}$ nats.

Predictor	MAE	Spearman	Precision	Recall	Accuracy
Predict zero	9.2	–	–	0.00	0.45
$\hat{I}_{\text{var}}\times$ median retained	8.2	0.74	0.81	0.83	0.80
Ridge (6 features)	10.1	0.64	0.61	0.93	0.63

TABLE XIX

Metadata added value under dataset shift (E33, exploratory; 10^{-3} nats). Image head and fusion model fitted on the source only (frozen ConvNeXt-T; harmonised age, sex, coarse site; complete cases) and applied unchanged to the target. $^{+}/^{-}$: 95% target-group bootstrap CI above/below 0. Diagonal (italic): the target's own cross-validated value with the same harmonised metadata. Melanoma and malignancy datasets are paired only within their label family. Differences in prevalence and label definition between datasets are not separated from metadata effects.

Source ↓ / target →	HAM10000	ISIC 2020	ISIC 2019 (non-HAM)		
HAM10000	2.1	1.1^{+}	-26.8^{-}		
ISIC 2020	15.9^{+}	0.8	11.3^{+}		
ISIC 2019 (non-HAM)	9.7^{+}	6.7^{+}	6.0		
Source ↓ / target →	**PAD-UFES-20**	**HIBA**	**MSK 2020**	**MILK10k (derm.)**	**MILK10k (clin.)**
PAD-UFES-20	2.9	−6.9	30.1^{+}	−2.8	-15.7^{-}
HIBA	-20.2^{-}	9.7	58.5^{+}	28.1^{+}	55.0^{+}
MSK 2020	-64.2^{-}	6.1	21.7	-146.6^{-}	-115.3^{-}
MILK10k (derm.)	-21.1^{-}	57.2^{+}	−3.5	12.3	62.7^{+}
MILK10k (clin.)	−7.7	56.0^{+}	20.6^{+}	25.9^{+}	9.8

TABLE XX

Whether to use metadata on a new dataset, decided from source datasets only (E55; leave one dataset out within a label family; frozen ConvNeXt-T, harmonised age, sex and site, complete cases; 10^{-3} nats). Δ: realised added value on the held-out target. Oracle: the better choice on the target. Source-validated: use if the mean leave-one-source-out added value among the sources is positive. Collection rule: standalone value of the pooled sources times their median retained fraction, use if $\geq 10^{-3}$ nats. Mean regret: never 9.5, always 3.6, source-validated 3.9, collection rule 3.6.

Target	n	Δ	Oracle	Source-val.	Coll. rule
HAM10000	9,761	5.43 [2.89, 8.10]	use	use	use
ISIC 2019	14,156	8.43 [3.43, 13.47]	use	use	use
ISIC 2020	32,531	2.12 [1.45, 2.76]	use	skip	use
PAD-UFES-20	1,089	−19.61 [−36.42, −2.41]	skip	use	use
HIBA	1,085	32.00 [21.03, 42.63]	use	use	use
MSK 2020	1,157	−5.65 [−13.69, 3.41]	skip	use	use
MILK10k (derm.)	5,086	18.79 [14.51, 23.18]	use	use	use

TABLE XXI

Mechanism checks for age erasure (E50, E52; ConvNeXt-T, fold assignment 0; 10^{-3} nats). Dose: erased coordinates mixed with independent noise, $c_t = \sqrt{1-t}\,c + \sqrt{t}\,\sigma\varepsilon$, so linear decodability falls with t; entries are erasure minus matched random subspaces at the same dose (†95% CI above 0). RF: rise in age's added value per unit rise in image-only log-loss, for erasure and for matched random subspaces (shown only where the denominator's CI excludes 0). Re-append: added value with the erased coordinate appended back, minus t=0. $\hat{a}$: cross-fitted image-derived age entered like a metadata variable after erasure, its added value minus the rise in age's added value (0 under pure redundancy). MLP R^2: held-out age decodability by a registered one-hidden-layer probe at $t = 0$ / after erasure / after one matched random subspace.

	Dose t (erasure − matched)				RF				
Image set	0.25	0.5	0.75	1	Erased	Matched	Re-append	$\hat{a}$ − rise	MLP R^2
ISIC 2020	0.23†	0.50†	0.80†	0.92†	0.71	0.10	0.15 [0.03, 0.27]	0.25 [−0.28, 0.77]	0.23 / 0.20 / 0.20
HIBA	−8.32	−8.84	−17.70	−53.55	0.37	0.52	49.84 [22.07, 77.07]	160.09 [100.97, 232.01]	0.11 / −1.14 / −2.53
MSK 2020	−2.44	−3.87	−9.19	−7.45	0.34	0.49	5.44 [−0.10, 10.57]	25.60 [11.54, 40.19]	0.02 / −0.76 / −2.65
HAM10000	1.56†	3.42†	6.08†	9.02†	0.79	0.08	0.82 [0.50, 1.13]	−0.04 [−3.37, 3.16]	0.28 / 0.21 / 0.20
ISIC 2019	1.32†	2.92†	4.75†	6.91†	0.97	0.03	1.91 [1.08, 2.75]	0.07 [−2.79, 3.33]	0.26 / 0.23 / 0.22
MILK10k (derm.)	0.12	−0.09	2.24	5.95†	0.60	0.35	2.86 [1.79, 3.96]	36.01 [27.04, 45.39]	0.18 / 0.05 / −0.15
PAD-UFES-20‡	0.10	0.03	0.44	0.31	–	–	0.42 [−0.86, 1.70]	2.96 [−1.31, 7.20]	−0.18 / −1.34 / −0.30
MILK10k (clin.)	0.32	2.54	4.03†	7.98†	0.65	0.38	5.36 [3.80, 6.98]	25.28 [17.18, 33.65]	0.24 / 0.13 / −0.12

TABLE XXII

Robustness of the age-erasure contrast (E50b, E50c, E53b; 10^{-3} nats, 95% grouped-bootstrap CIs). Fold assignments 1 and 2 and ResNet-18 use the primary arms only (erasure, nested matched random subspaces, shams). Full-feature fusion: both stages replaced by an L_2 logistic model on [features, age] with the same erasers. Class-conditional (exploratory, defined after the confirmatory results): the eraser removes the image's covariance with age minus its training-fold class mean, i.e. the age trace not explained by the label; it is applied without labels. Matching check as in Table IX.

Image set	Condition	Erased − matched	Erased − sham	Matching check
ISIC 2020	Fold assignment 1	1.08 [0.76, 1.38]	1.30 [0.94, 1.69]	−0.37 [−1.22, 0.47]
	Fold assignment 2	0.85 [0.53, 1.15]	1.20 [0.84, 1.55]	−1.73 [−2.61, −0.85]
	Full-feature fusion	1.49 [0.82, 2.20]	–	−0.30 [−1.11, 0.57]
	Class-conditional	0.86 [0.56, 1.16]	1.04 [0.73, 1.36]	0.08 [−0.74, 0.89]
HIBA	Class-conditional	61.34 [37.14, 85.10]	61.42 [37.22, 85.05]	283.45 [209.43, 366.67]
MSK 2020	Class-conditional	−3.43 [−12.02, 4.19]	−3.01 [−11.77, 4.94]	40.32 [23.43, 58.93]
HAM10000	Fold assignment 1	8.74 [6.11, 11.41]	10.69 [7.61, 13.78]	−8.95 [−13.33, −4.83]
	Fold assignment 2	8.65 [6.22, 11.22]	11.02 [8.10, 14.00]	−16.69 [−21.15, −12.59]
	ResNet-18	8.89 [6.37, 11.31]	11.63 [8.65, 14.72]	−9.67 [−14.52, −4.92]
	Full-feature fusion	8.12 [5.53, 10.75]	–	−12.88 [−17.35, −8.14]
	Class-conditional	4.05 [2.48, 5.65]	4.48 [2.78, 6.22]	−0.43 [−2.90, 1.96]
ISIC 2019	Fold assignment 1	6.74 [4.54, 8.97]	7.16 [4.75, 9.56]	−4.39 [−8.05, −0.44]
	Fold assignment 2	6.42 [4.21, 8.61]	7.19 [4.78, 9.57]	−6.37 [−9.89, −2.63]
	ResNet-18	6.22 [4.08, 8.37]	6.99 [4.63, 9.37]	−6.91 [−10.65, −2.76]
	Full-feature fusion	5.13 [3.66, 6.60]	–	−3.77 [−7.05, −0.41]
	Class-conditional	2.53 [1.43, 3.69]	2.74 [1.59, 3.91]	−0.34 [−1.86, 1.23]
MILK10k (derm.)	ResNet-18	22.19 [17.02, 27.22]	51.82 [44.06, 59.05]	6.87 [−0.65, 14.16]
	Full-feature fusion	23.49 [17.20, 29.94]	–	−39.64 [−48.14, −31.13]
	Class-conditional	13.62 [9.66, 17.36]	21.28 [16.60, 25.66]	2.32 [−4.09, 9.23]
PAD-UFES-20‡	ResNet-18	1.89 [−0.16, 4.11]	1.84 [−0.19, 3.99]	−4.53 [−9.76, 1.12]
	Class-conditional	−0.07 [−1.33, 1.23]	−0.04 [−1.22, 1.19]	−12.31 [−18.31, −6.33]
MILK10k (clin.)	Full-feature fusion	25.85 [19.79, 31.80]	–	−36.70 [−45.77, −27.95]
	Class-conditional	14.05 [9.88, 18.27]	25.24 [20.53, 30.03]	1.50 [−6.28, 9.61]

TABLE XXIII

Attribute specificity (E50, E51; ConvNeXt-T, fold assignment 0; 10^{-3} nats, 95% grouped-bootstrap CIs). Each entry is erasure minus performance-matched random subspaces, in the added value of age or of site. Registered rule: an eraser is selective if its effect on its own variable and the paired difference between its effect on its own and on the other variable both have CIs above 0. HIBA (site) was the confirmatory test; HAM10000 a replication; MILK10k (derm.) and PAD-UFES-20 exploratory. Site erasure's matching check failed on HIBA and PAD-UFES-20 (matched control far more damaging) and passed or was conservative on HAM10000 and MILK10k.

Image set	Eraser	Age's added value	Site's added value	Own − other variable	Selective
HAM10000	Age	9.05 [6.49, 11.50]	0.39 [−0.97, 1.69]	8.66 [5.84, 11.32]	✓
	Site	0.55 [−0.49, 1.61]	4.26 [1.64, 6.82]	3.71 [1.01, 6.38]	✓
MILK10k (derm.)	Age	9.53 [4.42, 14.54]	−1.47 [−2.65, −0.25]	10.99 [5.77, 15.91]	✓
	Site	−1.15 [−2.48, 0.19]	3.83 [1.37, 6.31]	4.98 [2.44, 7.55]	✓
HIBA	Age	−116.56 [−161.66, −79.36]	−38.07 [−49.14, −26.37]	−78.49 [−126.23, −39.28]	×
	Site	−128.01 [−154.19, −103.34]	−54.11 [−82.55, −27.93]	73.90 [39.91, 106.69]	×
PAD-UFES-20[‡]	Age	0.22 [−2.53, 2.93]	−1.91 [−4.05, −0.08]	2.12 [−0.74, 5.20]	×
	Site	−0.93 [−3.42, 1.57]	−87.07 [−105.46, −69.29]	−86.14 [−104.37, −68.31]	×

TABLE XXIV

Added value of all metadata at ConvNeXt-T under four fusion estimators (E53; 10^{-3} nats, 95% grouped-bootstrap CIs; strictly cross-fitted, same folds, groups and test images; all images). (a) Score fusion: logistic second stage on the image score, as in the rest of the paper. (b) Full-feature fusion: L_2 logistic model on [image features, metadata], penalty by grouped inner CV. (c) Boosted fusion: gradient-boosted trees on [image score, raw metadata], fixed hyperparameters. (d) Image score plus missingness indicators only. H1: $\Delta_{\text{HSV}} > \Delta_{\text{ConvNeXt}}$ (paired, one-sided $p < 0.05$) under (a) (b) (c).

Image set	(a) Score	(b) Full-feature	(c) Boosted	(d) Missingness only	H1 (a)(b)(c)
ISIC 2020	0.92 [0.11, 1.68]	0.87 [0.14, 1.54]	1.36 [0.49, 2.23]	−0.03 [−0.09, 0.02]	✓ ✓ ✓
HIBA	3.80 [−12.63, 20.13]	7.03 [2.32, 11.88]	3.64 [−14.05, 20.38]	−12.82 [−24.01, −3.38]	✓ ✓ ✓
MSK 2020	17.40 [6.89, 27.94]	1.66 [−5.89, 8.86]	15.17 [1.27, 29.02]	1.32 [0.60, 2.19]	✓ ✓ ✓
HAM10000	2.38 [−2.13, 6.19]	4.96 [2.12, 7.90]	6.83 [2.46, 11.21]	−0.69 [−3.42, 1.28]	✓ ✓ ✓
ISIC 2019	9.80 [3.93, 16.08]	8.72 [5.36, 12.26]	16.01 [9.02, 23.36]	2.49 [0.94, 3.89]	✓ ✓ ✓
MILK10k (derm.)	8.29 [−0.74, 15.53]	11.99 [6.62, 17.21]	15.69 [10.50, 20.84]	−0.22 [−0.86, 0.38]	✓ ✓ ✓
PAD-UFES-20	206.87 [179.27, 232.60]	181.47 [160.46, 202.24]	224.75 [196.95, 250.62]	158.31 [137.94, 179.19]	✓ ✓ ✓
MILK10k (clin.)	6.30 [−2.25, 13.03]	7.56 [1.89, 12.97]	12.32 [7.69, 16.98]	0.11 [−0.60, 0.81]	✓ ✓ ✓

TABLE XXV

Robustness of H1 and H2 (✓: holds; ×: fails; –: not derivable). Registered and strictly cross-fitted rows: Holm-adjusted within the registered waves (HAM10000 unadjusted). Alternative label: malignant vs. benign for HAM10000 and ISIC 2019, melanoma vs. rest elsewhere; ISIC 2020 has no distinct alternative. Alternative-label, complete-case and encoding rows use the strictly cross-fitted estimator and uncorrected one-sided tests ($p < 0.05$) or point estimates as stated; fold assignment 0 unless stated.

Condition	HAM	ISIC20	ISIC19	PAD	HIBA	MSK	MILK-d	MILK-c
H1: $\Delta_{\text{HSV}} > \Delta_{\text{ConvNeXt}}$								
Original estimator (registered)	✓	✓	✓	✓	✓	✓	✓	✓
Strictly cross-fitted	✓	✓	✓	✓	✓	✓	✓	✓
Fold assignments 0/1/2 (point > 0)	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
Group-weighted (point > 0)	✓	✓	✓	✓	✓	✓	✓	✓
Alternative label	✓	–	✓	✓	✓	✓	×	×
Complete cases	✓	✓	✓	✓	✓	✓	✓	✓
No missing indicators (point > 0)	✓	✓	✓	✓	✓	✓	✓	✓
H2: $\Delta_{\text{ResNet-18}} > \Delta_{\text{ConvNeXt}}$								
Original estimator (registered)	✓	×	×	✓	✓	×	✓	✓
Strictly cross-fitted	✓	✓	×	✓	✓	×	✓	✓
Fold assignments 0/1/2 (point > 0)	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3
Group-weighted (point > 0)	✓	✓	×	✓	✓	✓	✓	✓
Alternative label	✓	–	✓	✓	×	×	×	×
Complete cases	✓	✓	×	×	✓	✓	✓	✓
No missing indicators (point > 0)	–	–	–	–	–	–	–	–

TABLE XXVI

Where metadata help (ConvNeXt-T unless stated; strictly cross-fitted, fold assignment 0). ρ_{LE}, ρ_h: Spearman correlation of realised per-image benefit with local label entropy on the inductive graph (pre-registered H1, 95% grouped CI) and with the image model's predictive entropy (H2). H4 p: one-sided, unadjusted, graph statistics vs. model-based baselines. ρ_K, ρ_G: post hoc rank correlation of benefit with the fusion model's expected gain K_i and with graph-local benefit. Top 25%: share of total benefit captured by the quarter of images with the largest K_i (HIBA omitted: total benefit near zero). E: normalised Dirichlet energy of benefit relative to the label-by-uncertainty stratified null (HSV, ConvNeXt-T). s_H: share of the HSV→ConvNeXt-T log-attenuation due to image uncertainty (exploratory).

Image set	ρ_{LE}	ρ_h	H4 p	ρ_K	ρ_G	Top 25%	E_{HSV}	E_{CNX}	s_H
HAM10000	0.03 [−0.00, 0.05]	0.09	0.822	0.13	0.11	90%	0.986	0.997	0.13
ISIC 2020	0.11 [0.08, 0.13]	0.16	0.014	0.15	0.19	97%	0.986	0.999	0.09
ISIC 2019 (non-HAM)	0.10 [0.07, 0.13]	0.17	0.281	0.21	0.15	105%	0.976	0.989	0.33
PAD-UFES-20	0.26 [0.22, 0.30]	0.53	0.943	0.69	0.13	70%	1.003	1.002	0.61
HIBA	0.02 [−0.04, 0.09]	0.13	0.988	0.15	−0.02	–	0.979	1.005	0.13
MSK 2020	−0.00 [−0.06, 0.06]	0.09	0.011	0.18	0.15	86%	1.002	0.994	0.12
MILK10k (derm.)	0.12 [0.09, 0.15]	0.17	0.003	0.18	0.12	82%	0.989	0.995	0.13
MILK10k (clin.)	0.16 [0.13, 0.18]	0.17	0.373	0.16	0.12	76%	0.973	0.997	0.11

TABLE XXVII
Share of images whose melanoma-evidence map overlaps its lesion mask significantly ($p \leq 0.05$) under the circular-shift null and under the mask-swap null (masks of 199 other images in the same lesion-area decile), for real class-activation maps and for a fixed centred Gaussian map that carries no image information (E48; HAM10000).

		Real maps		Centred Gaussian	
Backbone	Images	Shift	Swap	Shift	Swap
ResNet-18	2,225	55%	16%	88%	5%
ResNet-50	2,225	37%	10%	88%	5%
ConvNeXt-T	2,225	57%	19%	88%	5%

TABLE XXVIII
Class-activation evidence vs. expert lesion masks (HAM10000, ResNet-50, 2225 images). Enrichment: share of positive evidence inside the lesion divided by the lesion's area share. Significance at $p \leq 0.05$.

Evidence map	Enrichment > 1	Shift null	Pixel perm.
Predicted class	32%	10%	30%
Melanoma (all images)	84%	37%	82%

TABLE XXIX
Melanoma evidence (exact CAM) vs. expert lesion masks across backbones (HAM10000; all melanomas plus a matched sample of other lesions). Significance at $p \leq 0.05$: exact circular-shift null vs. pixel permutation. Last column: median lesion area fraction of detected / missed melanomas.

Backbone	Images	Enrich.> 1	Shift null	Pixel perm.	Area det./miss.
ResNet-18	2,222	90%	53%	89%	0.49 / 0.37
ResNet-50	2,222	84%	36%	82%	0.52 / 0.37
ConvNeXt-T	2,222	91%	56%	90%	0.49 / 0.33

References

[1] J. Höhn et al., "Integrating patient data into skin cancer classification using convolutional neural networks: Systematic review," Journal of Medical Internet Research, vol. 23, no. 7, p. e20708, 2021.

[2] A. G. C. Pacheco and R. A. Krohling, "The impact of patient clinical information on automated skin cancer detection," Computers in Biology and Medicine, vol. 116, p. 103545, 2020.

[3] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," Scientific Data, vol. 5, p. 180161, 2018.

[4] V. Rotemberg et al., "A patient-centric dataset of images and metadata for identifying melanomas using clinical context," Scientific Data, vol. 8, 2021, arXiv:2008.07360.

[5] M. Combalia et al., "BCN20000: Dermoscopic lesions in the wild," Scientific Data, 2024.

[6] A. G. C. Pacheco et al., "PAD-UFES-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones," Data in Brief, vol. 32, p. 106221, 2020.

[7] M. A. Ricci Lara, M. V. Rodríguez Kowalczuk, M. L. Eliceche, M. G. Ferraresso, D. R. Luna, S. E. Benitez, and L. D. Mazzuoccolo, "A dataset of skin lesion images collected in Argentina for the evaluation of AI tools in this population," Scientific Data, vol. 10, p. 712, 2023.

[8] Memorial Sloan Kettering Cancer Center, "Consecutive biopsies for melanoma across year 2020," ISIC Archive collection, https://doi.org/10.34970/151324, 2020, accessed October 2026.

[9] P. Tschandl, B. N. Akay, C. Rosendahl, V. Rotemberg, V. Todorovska, J. Weber, A. K. Wolber, C. Müller, N. Kurtansky, A. Halpern, W. Weninger, and H. Kittler, "MILK10k: A hierarchical multimodal imaging-learning toolkit for diagnosing pigmented and nonpigmented skin cancer and its simulators," Journal of Investigative Dermatology, vol. 146, no. 2, pp. 357–364.e7, 2026.

[10] S. Ahammed, X. Cui, W. Lu, and M. H. Yap, "Skin lesion classification using dermoscopic images and clinical metadata: Insights from multimodal models," in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR) Workshops, 2025, pp. 222–230.

[11] C. Müller, A. Wolber, A. Lallas, G. Argenziano, I. Zalaudek, H. P. Soyer, S. Puig, B. N. Akay, A. M. Forsea, J. Malvehy, A. Marghoob, Z. Apalla, C. Rosendahl, K. Liopyris, O. Reiter, C. Longo, P. Tschandl, and H. Kittler, "Comparative and complementary diagnostic value of dermatoscopy and clinical close-up photography in skin cancer diagnosis: A study from the MILK10k dataset," Journal of the American Academy of Dermatology, vol. 95, no. 1, pp. 70–77, 2026.

[12] O. V. Demler, M. J. Pencina, and R. B. D'Agostino, "Misuse of DeLong test to compare AUCs for nested models," Statistics in Medicine, vol. 31, no. 23, pp. 2577–2587, 2012.

[13] M. H. Gail, S. Wieand, and S. Piantadosi, "Biased estimates of treatment effect in randomized experiments with nonlinear regressions and omitted covariates," Biometrika, vol. 71, no. 3, pp. 431–444, 1984.

[14] I. C. Covert, W. Qiu, M. Lu, N. Y. Kim, N. J. White, and S.-I. Lee, "Learning to maximize mutual information for dynamic feature selection," in Proceedings of the 40th International Conference on Machine Learning, ser. PMLR, vol. 202, 2023, pp. 6424–6447.

[15] S. Gadgil, I. Covert, and S.-I. Lee, "Estimating conditional mutual information for dynamic feature selection," in International Conference on Learning Representations (ICLR), 2024.

[16] N. Belrose, D. Schneider-Joseph, S. Ravfogel, R. Cotterell, E. Raff, and S. Biderman, "LEACE: Perfect linear concept erasure in closed form," in Advances in Neural Information Processing Systems (NeurIPS), vol. 36, 2023.

[17] Y. Elazar, S. Ravfogel, A. Jacovi, and Y. Goldberg, "Amnesic probing: Behavioral explanation with amnesic counterfactuals," Transactions of the Association for Computational Linguistics, vol. 9, pp. 160–175, 2021.

[18] A. Kumar, C. Tan, and A. Sharma, "Probing classifiers are unreliable for concept removal and detection," in Advances in Neural Information Processing Systems (NeurIPS), vol. 35, 2022, pp. 17 994–18 008.

[19] E. Yagis et al., "Effect of data leakage in brain MRI classification using 2D convolutional neural networks," Scientific Reports, vol. 11, 2021.

[20] A. F. Alexander-Bloch et al., "On testing for spatial correspondence between maps of human brain structure and function," NeuroImage, 2018.

[21] R. D. Markello et al., "neuromaps: structural and functional interpretation of brain maps," Nature Methods, 2022.

[22] T. M. Cover and J. A. Thomas, Elements of Information Theory, 2nd ed. Wiley, 2006.

[23] S. Bates, T. Hastie, and R. Tibshirani, "Cross-validation: What does it estimate and how well does it do it?" Journal of the American Statistical Association, vol. 119, no. 546, pp. 1434–1445, 2024.

[24] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770–778.

[25] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2022, pp. 11 976–11 986.

[26] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in Proc. Int. Conf. Learning Representations (ICLR), 2021.

[27] S. Holm, "A simple sequentially rejective multiple test procedure," Scandinavian Journal of Statistics, vol. 6, no. 2, pp. 65–70, 1979.

[28] M. Singh, L. Gustafson, A. Adcock, V. de Freitas Reis, B. Gedik, R. P. Kosaraju, D. Mahajan, R. Girshick, P. Dollár, and L. van der Maaten, "Revisiting weakly supervised pre-training of visual perception models," in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2022, pp. 804–814.

[29] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," in Advances in Neural Information Processing Systems (NeurIPS), vol. 31, 2018, pp. 9505–9515.

[30] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 2921–2929.